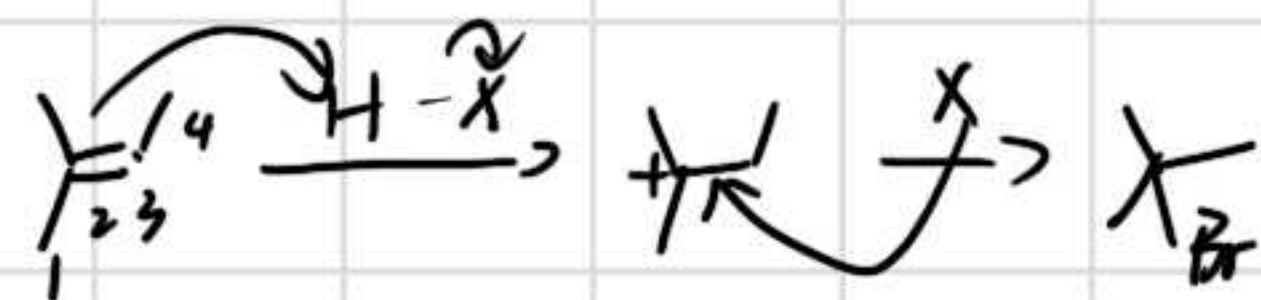


烯

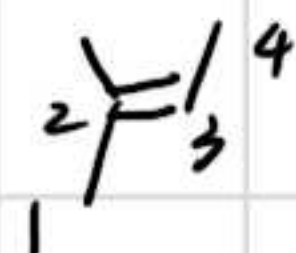
亲电加成: ① + HX

C(C<sup>+</sup>中间体)  
区域: 由C<sup>+</sup>稳定性决定  
Br到H少处

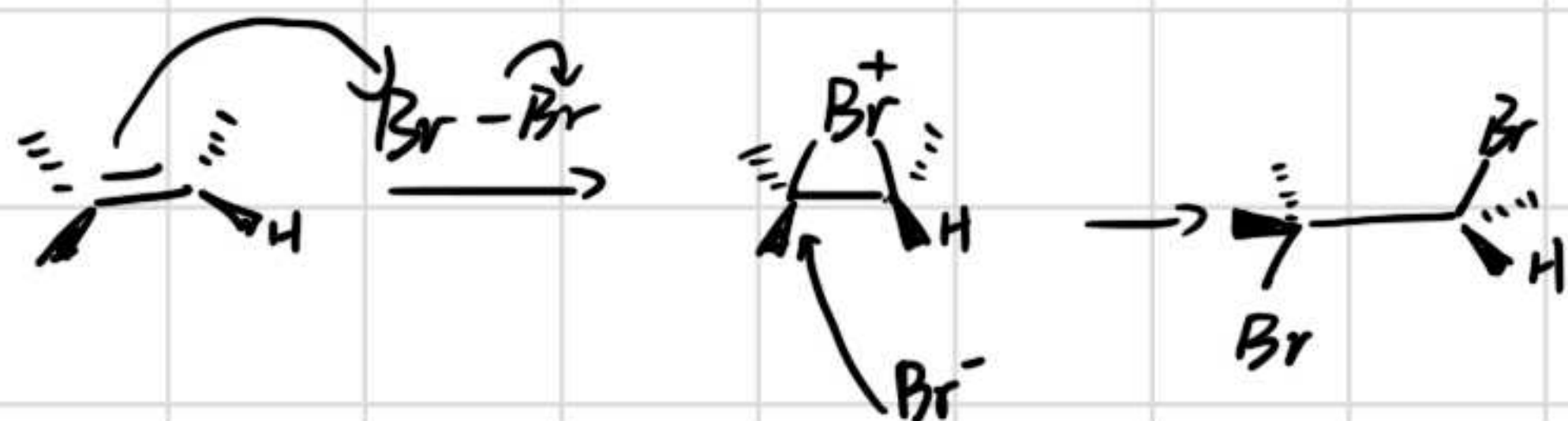


\* ROOR, HX

过氧化物: 自由基机理  
自由基稳定性决定, Br到H多处

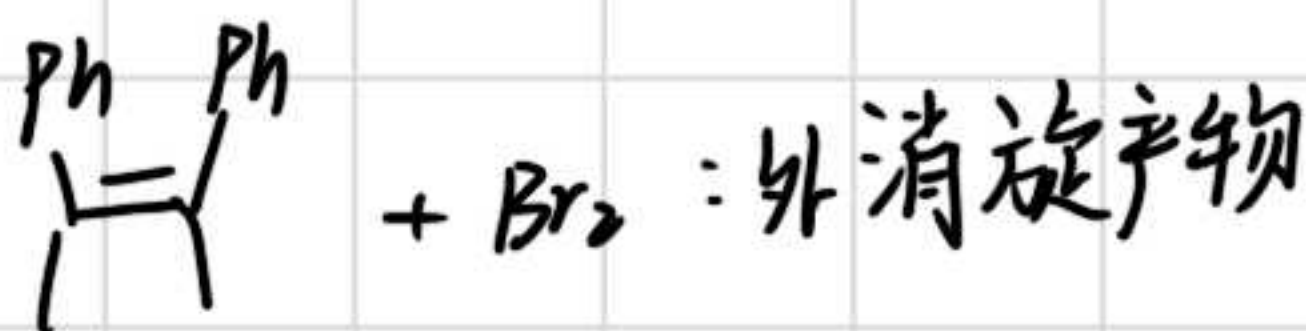
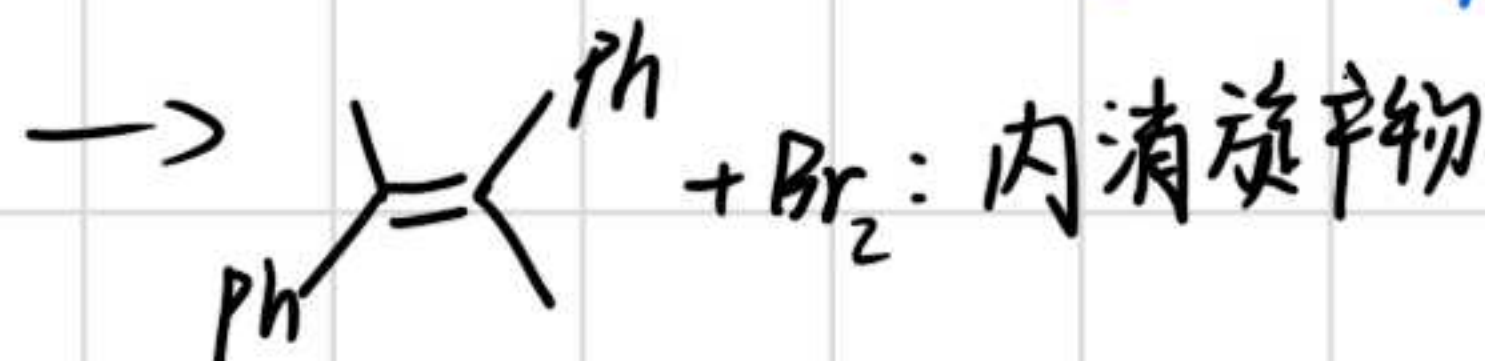


② 与X<sub>2</sub>

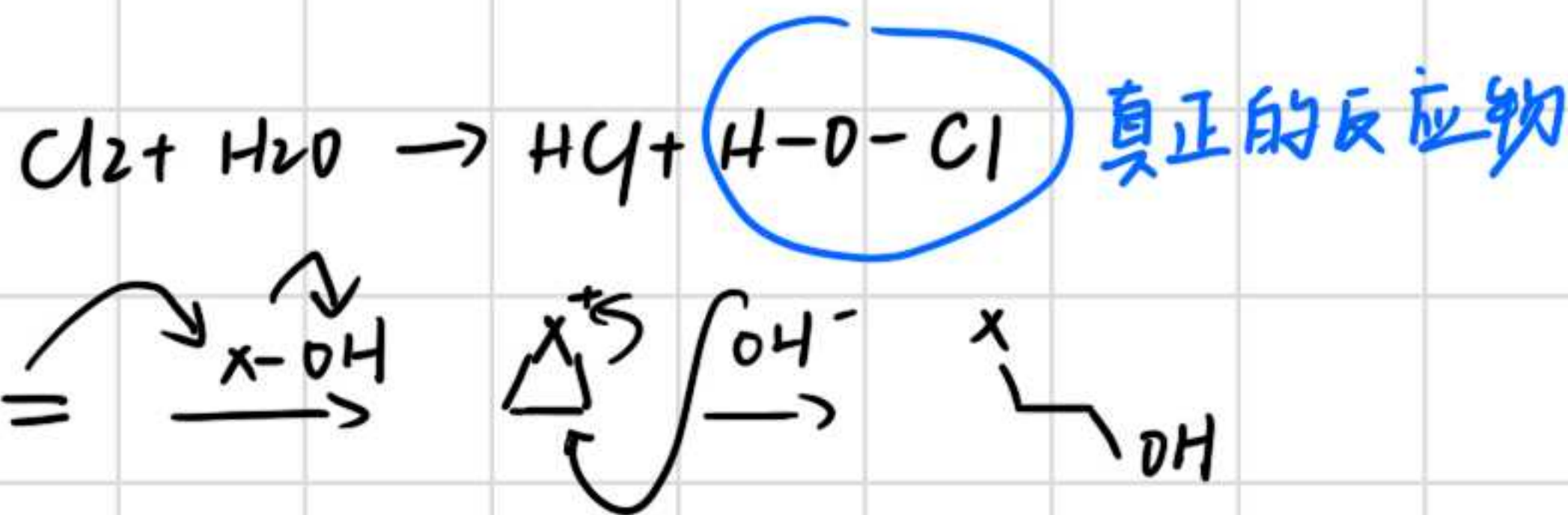


区域: Br<sup>-</sup>会去进攻C<sup>+</sup>更稳定的那个C  
三元环的2根键不等长, 类似于  $\text{C}^+-\text{Br}$

立体: 反式共面加成



\* 与X<sub>2</sub>反应, 水/醇作溶剂







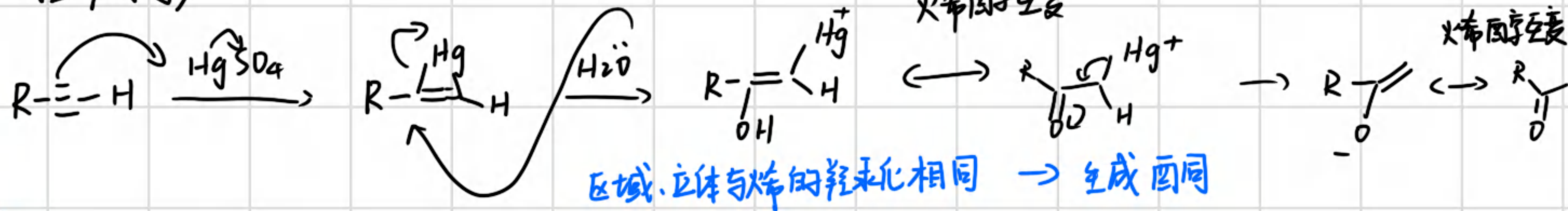


炔烃水合 (生成 醛/酮)

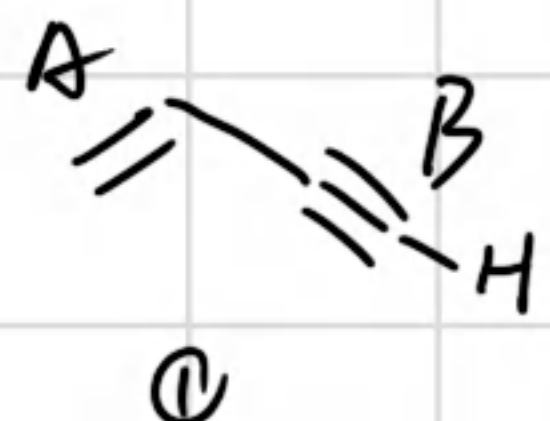
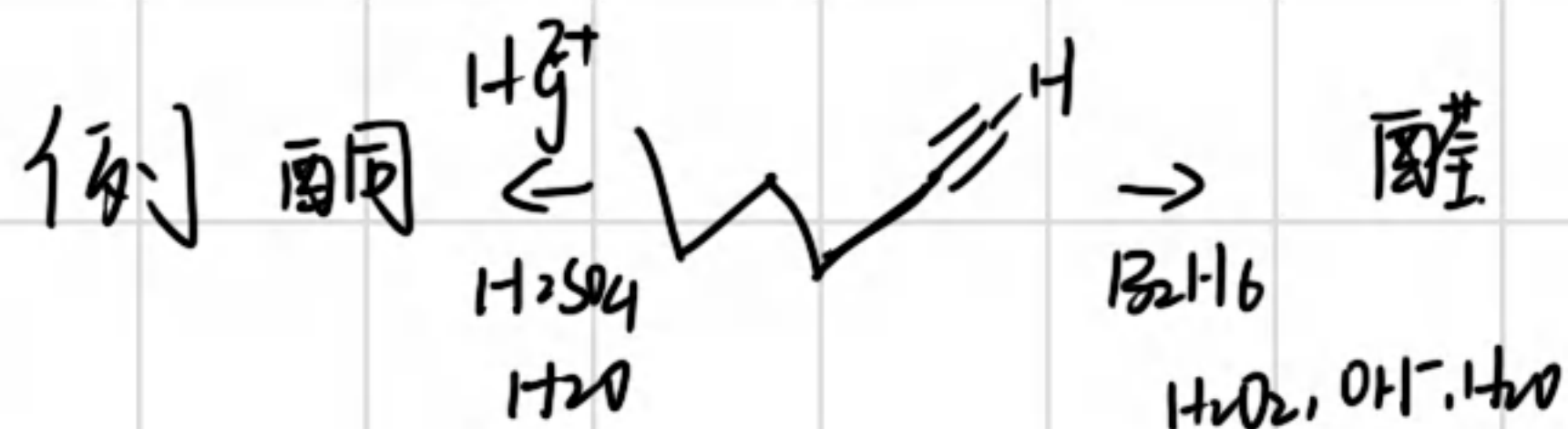
羟汞化:  $H_2SO_4, H_2O$

催化量  $HgSO_4$

炔烃羟汞化中  
 $HgCOAc_2$  是一当量的

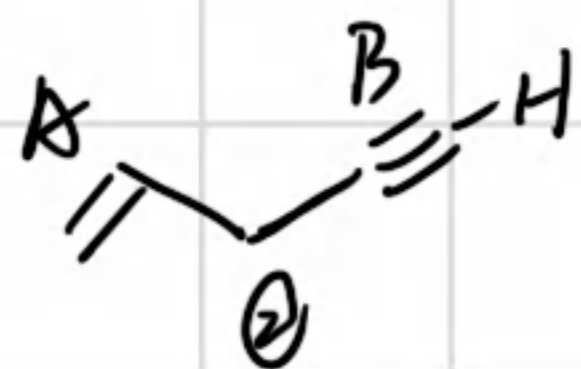


硼氢化-氧化 区域、立体与烯的硼氢化氧化相同  $\rightarrow$  可能生成醛 (边缘炔烃)



哪优先羟汞化?

①: B



②: A

D-A 反应

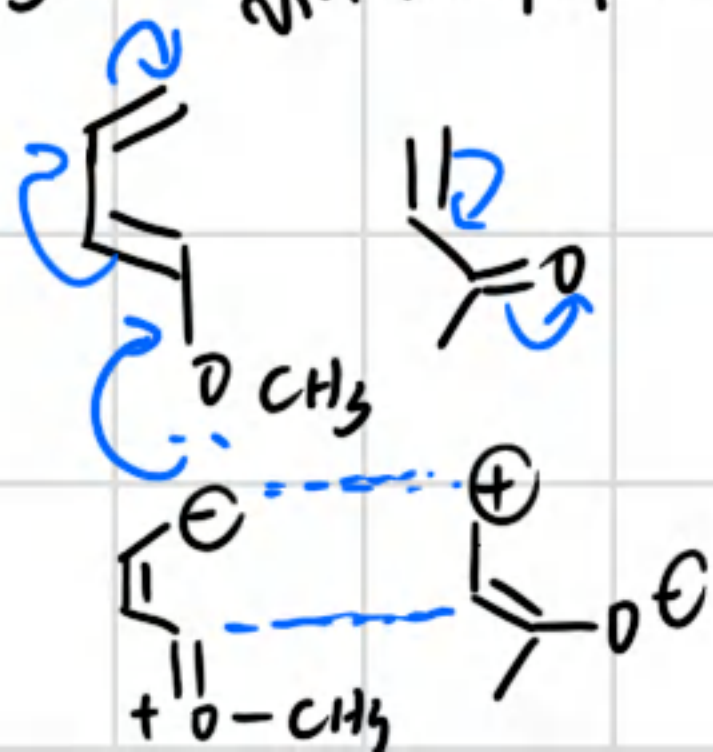
双烯体 (富  $e^-$  有利)

亲双烯体 (缺  $e^-$  有利)

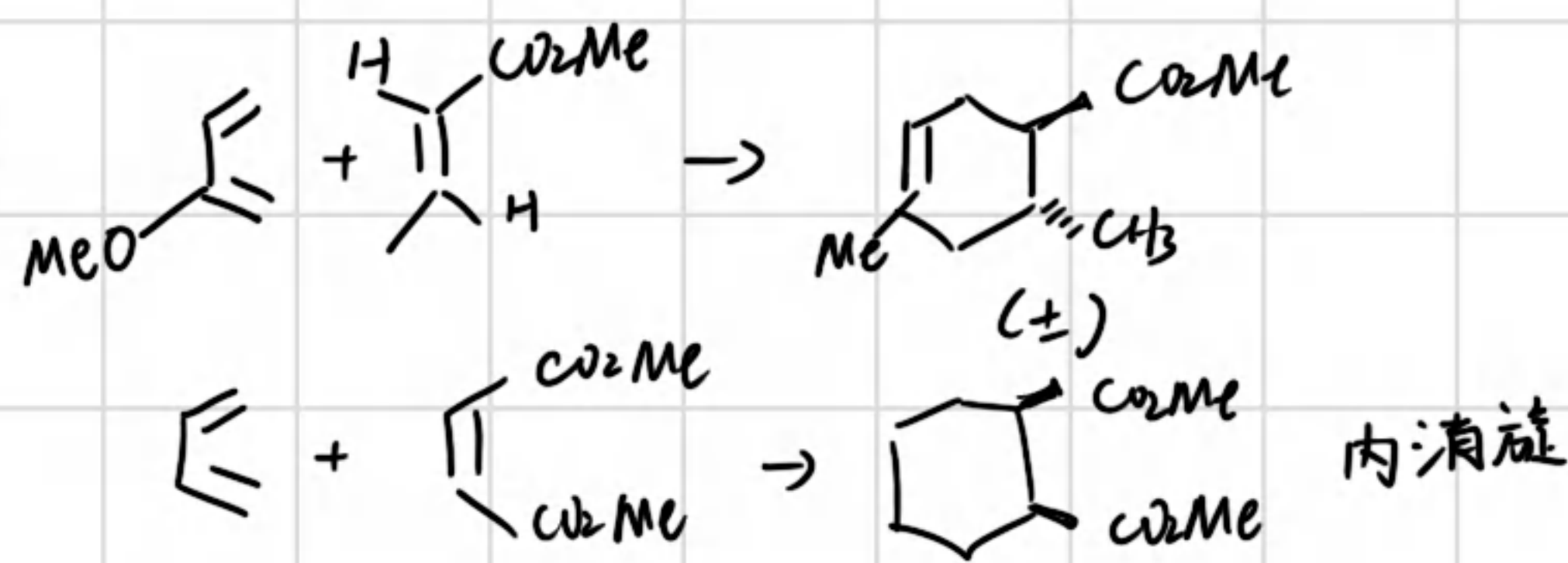
区域:  $\delta^+$  与  $\delta^-$  对上

立体: 构型保持

eg.



eg. 立体

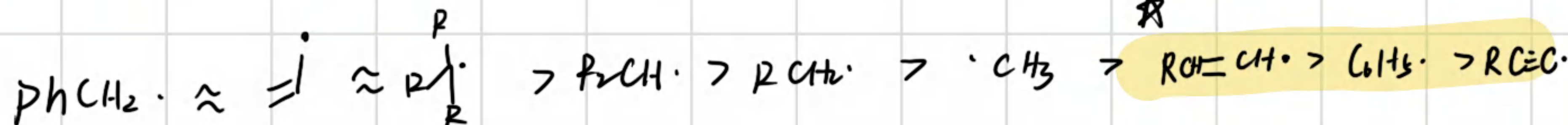
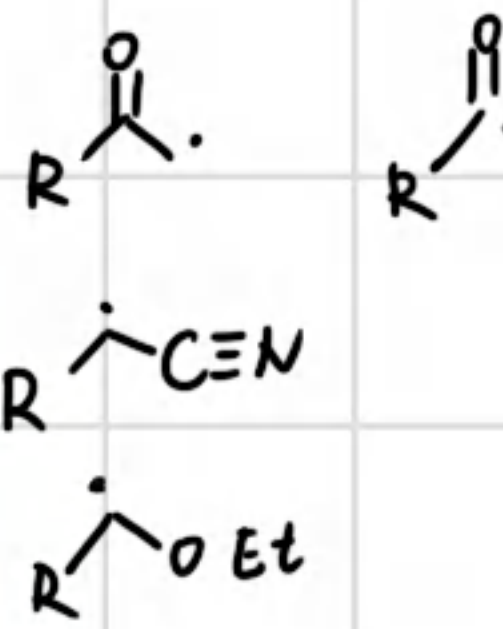




自由基: 区域选择性: 自由基中间体要稳定

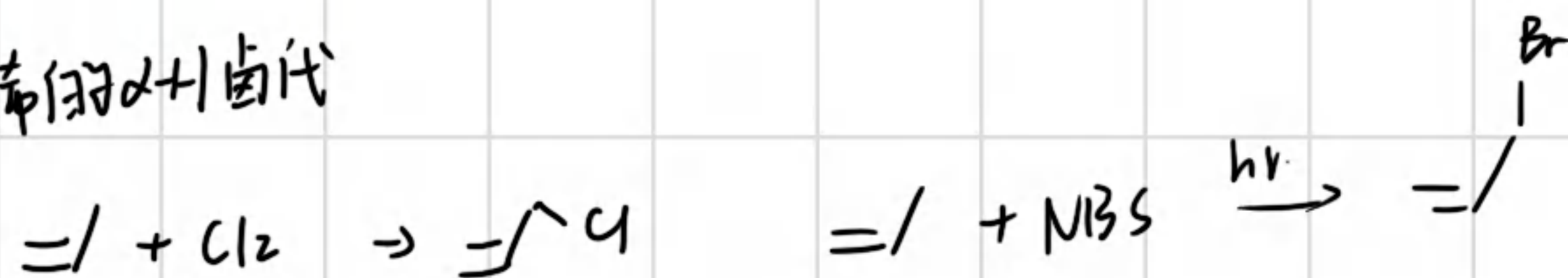
↓  
机理不要求, 特别关注  
溴化反应区域选择性  
性伏于氯化

- ①. 附近位阻大
- ②. 周围有给/吸e-基均可
- ③. 中间原子电负性小会稳定  
(I· > Br· > Cl· > F· > H·)



条件: 不与其他反应弄混  
hv / 500°C / Δ / 高温 /  
(PhCOO)<sub>2</sub> 等过氧化物

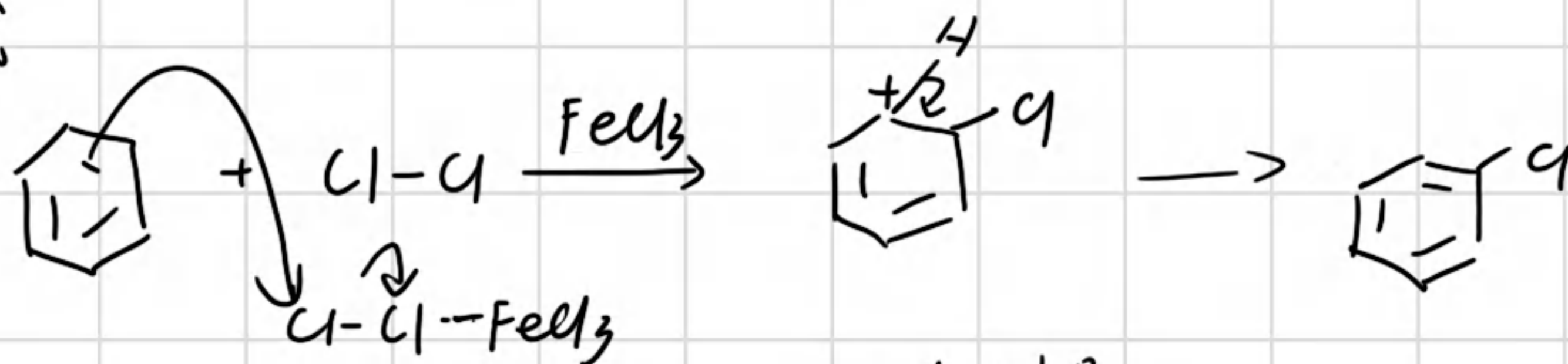
烯的α+1卤代



烯与HX加成 (见“烯”部分)

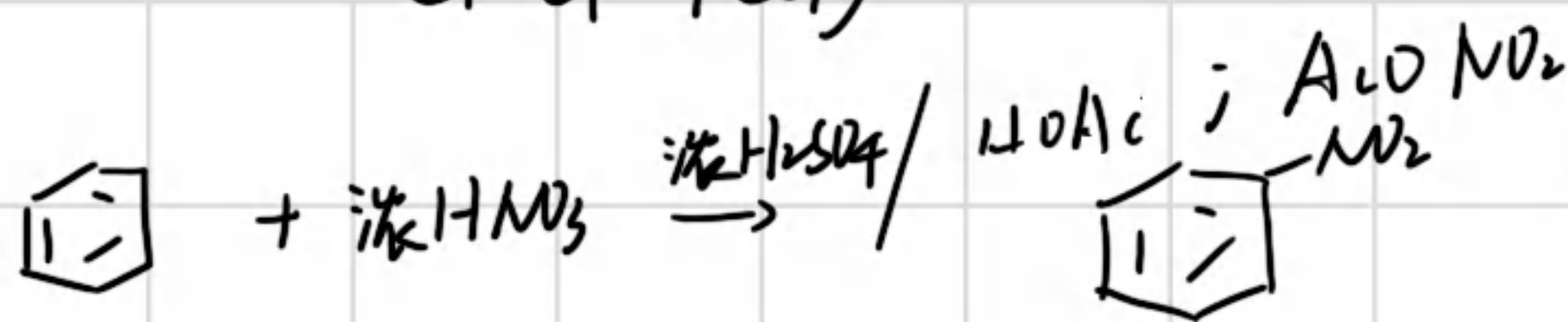
芳香亲电取代

卤化

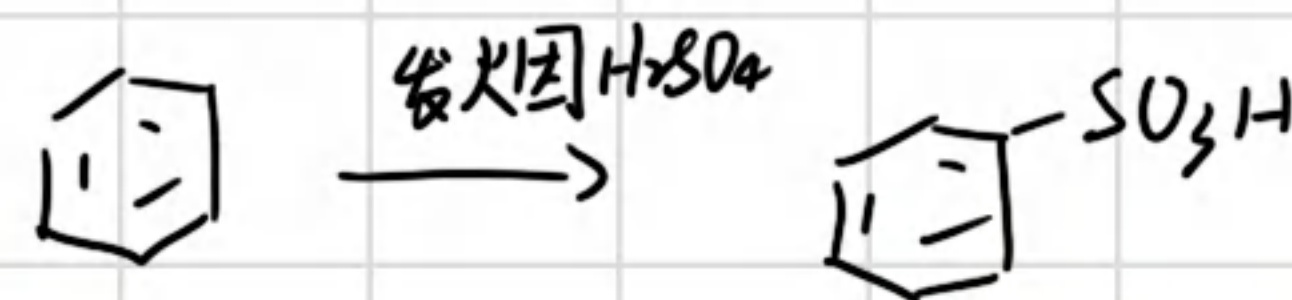


机理: 先加成后消除

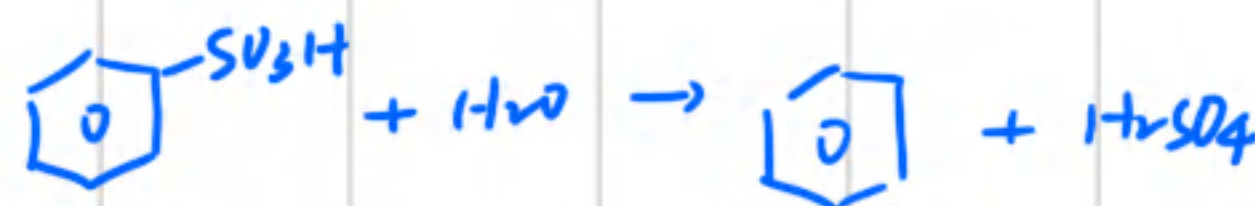
硝化



也可发生磺化



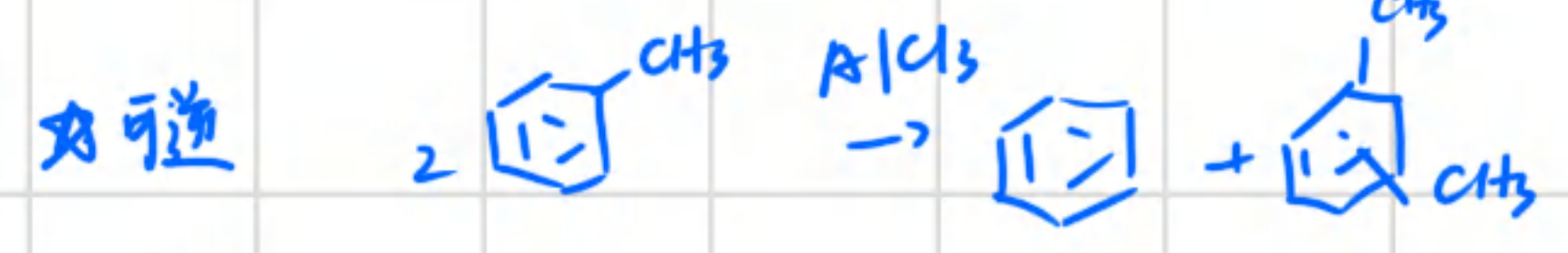
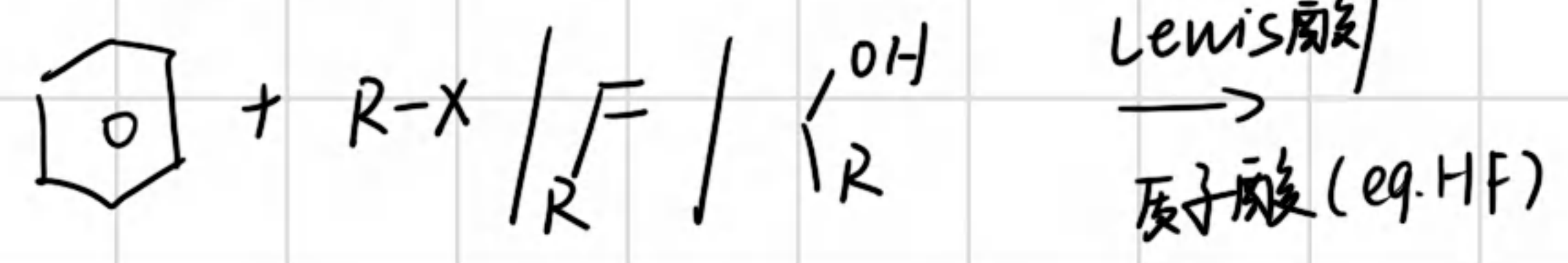
★ 可逆



但无法自-克烷/酮基化

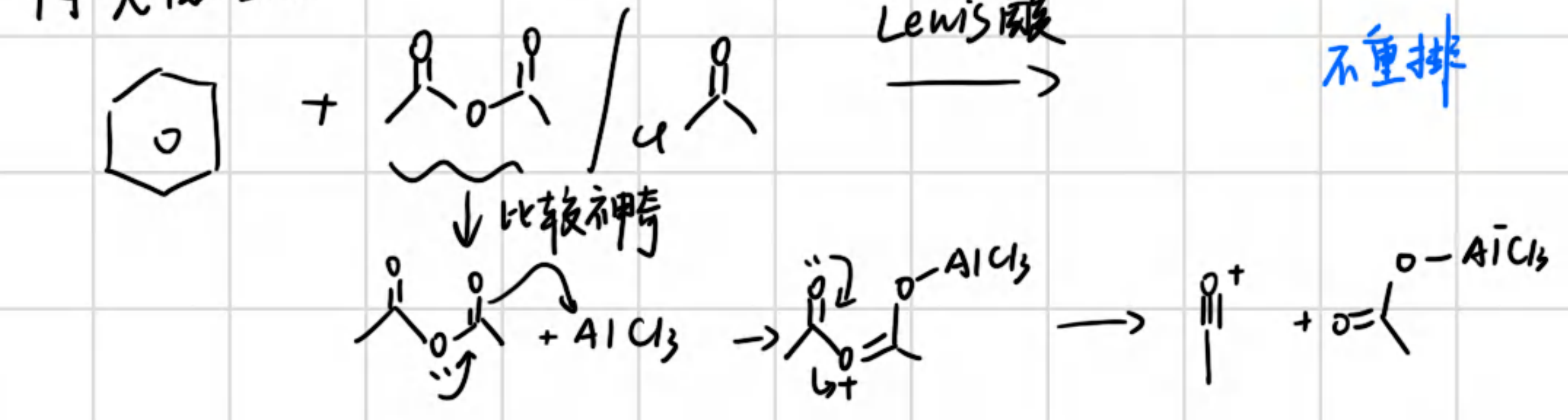


付-克烷基化



★ 副:  $C^+$  重排 多烷基化

付-克酰基化



不重排

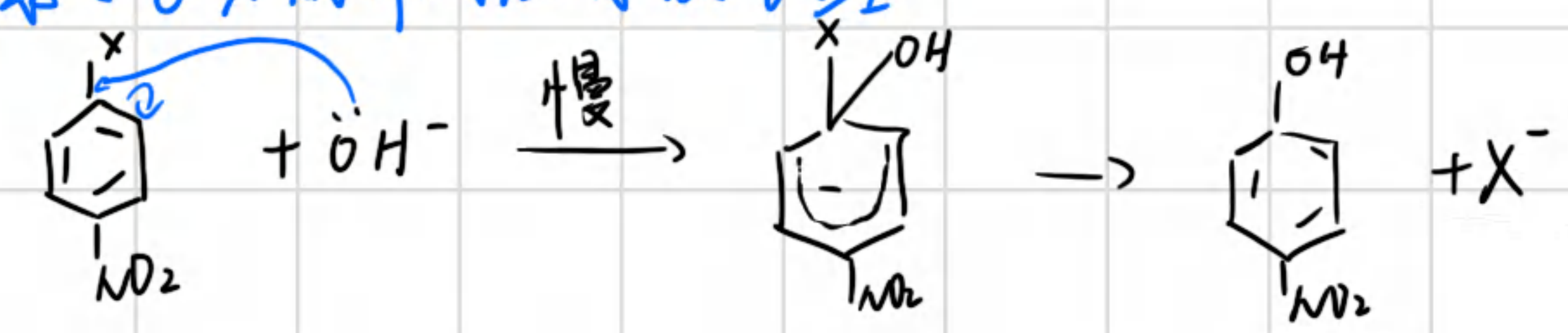
★ 苯环连了  $NR_3, NO_2, COOH, CO_2R, SO_3H$  等

c1ccccc1[NH2]  $\times \Rightarrow$   $-NH_2$  与 Lewis 酸成盐, 使  $-NH_2$  吸  $e^-$  基, 苯环电子云 ↓ 也不行!

芳香亲核取代

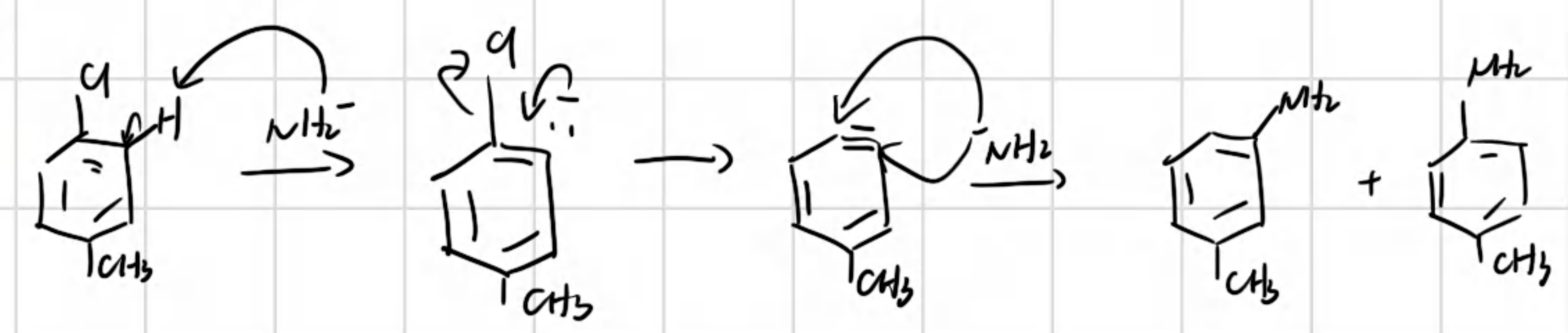
①. 加成-消除机理

要求:  $C-X$  的邻对位有吸电子基



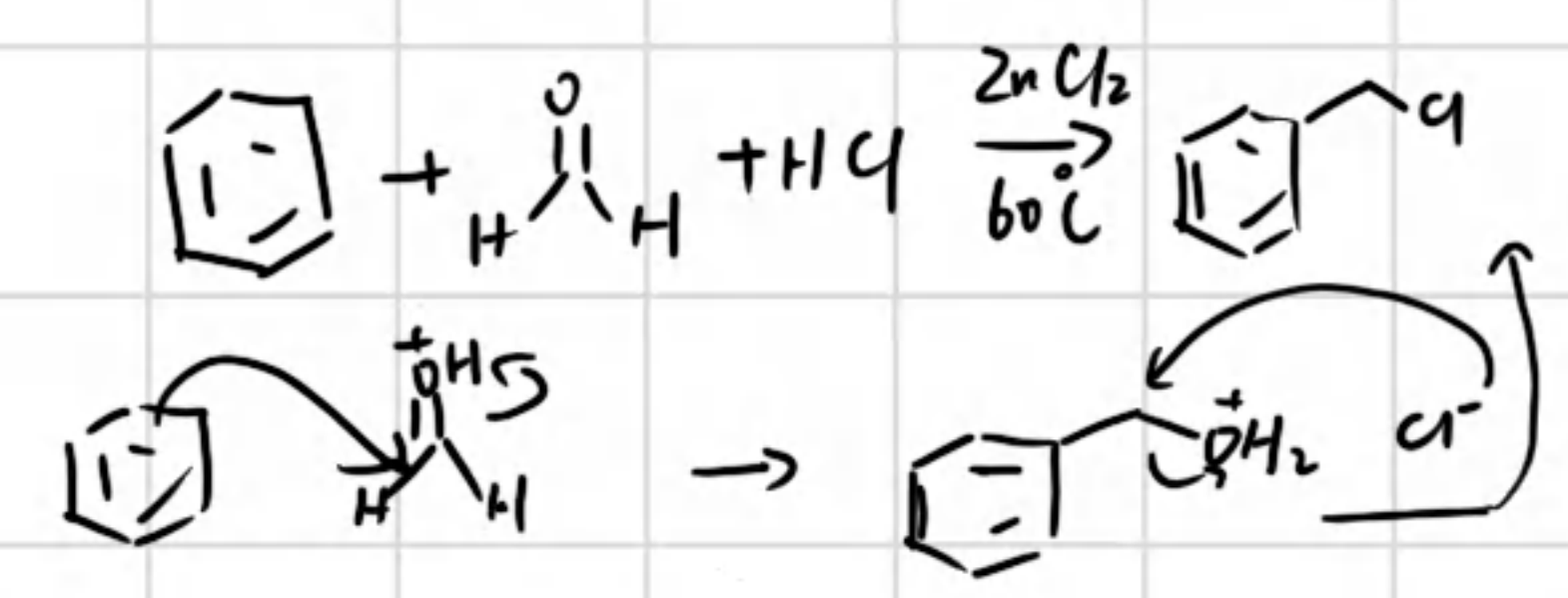
②. 苯炔机理 (消除-加成)

要求: 强碱 ( $KNH_2, NH_3$ )



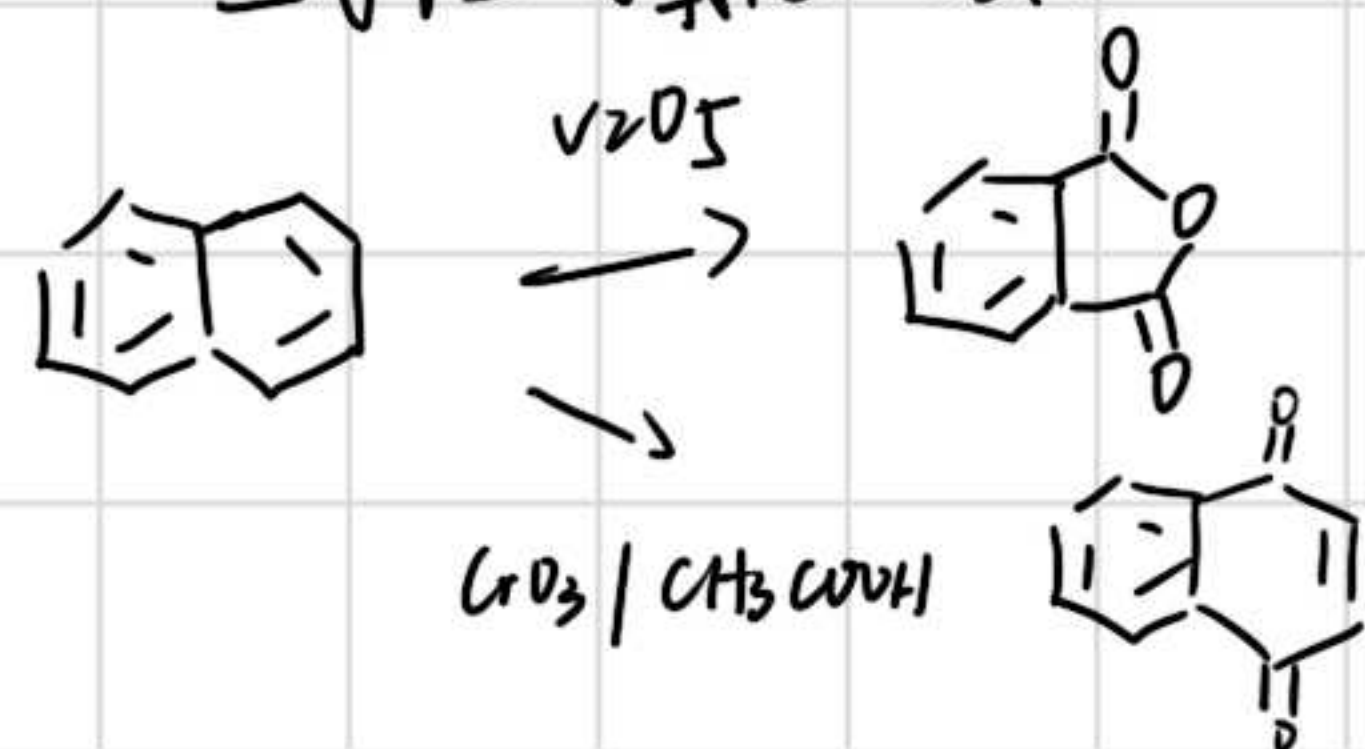
该活泼中间体可用 D-A 反应捕获

★ 氯甲基化反应

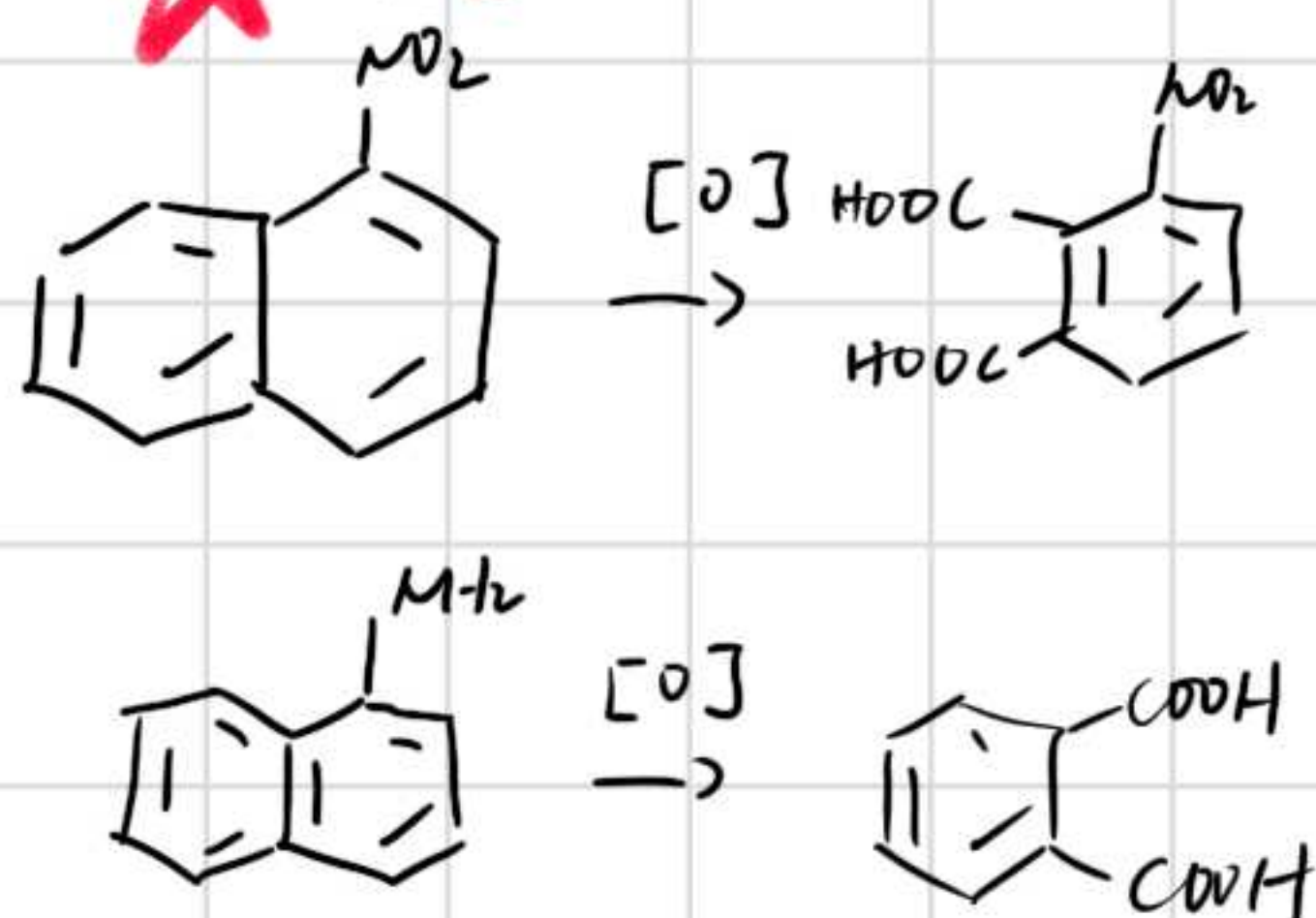




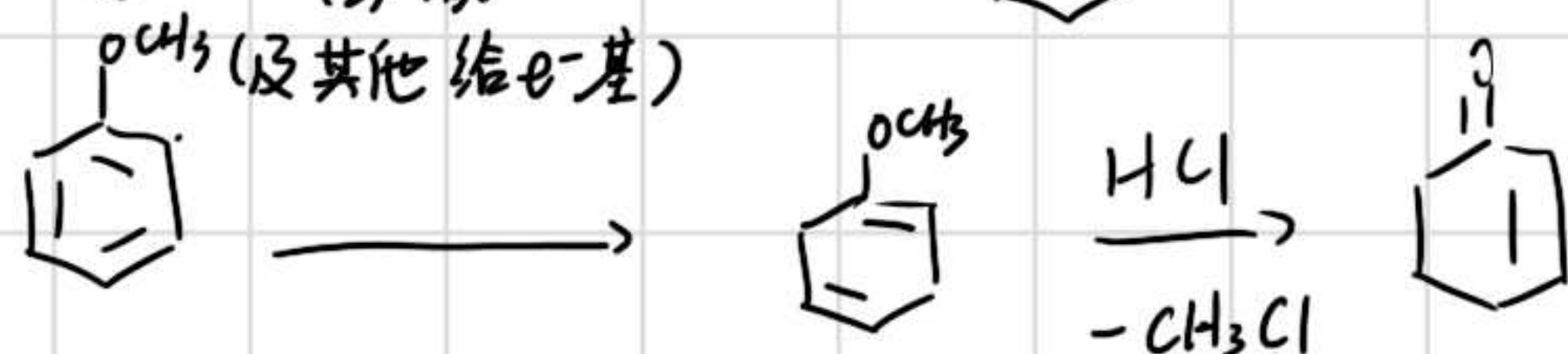
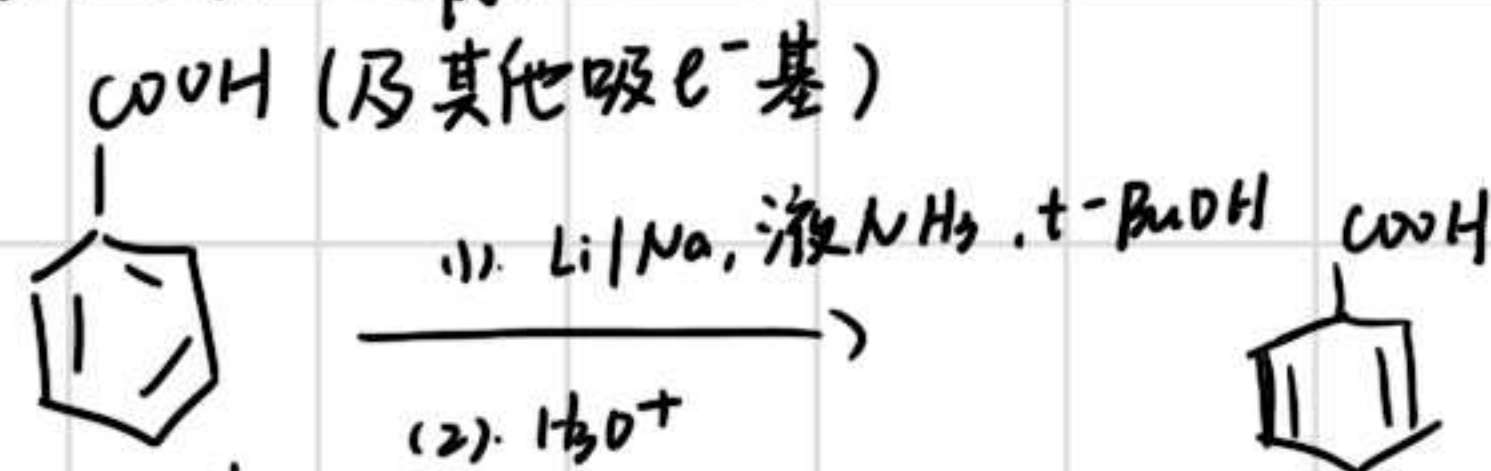
\* 一些奇怪的氧化 (去芳构化)



★ 富电子的易被氧化



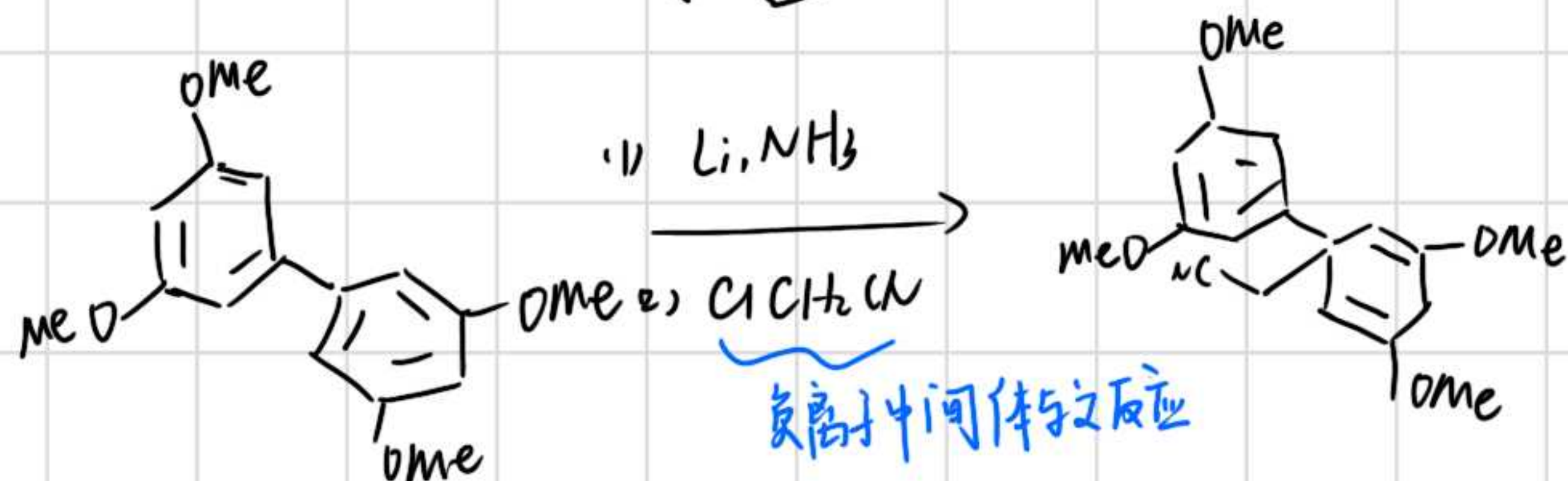
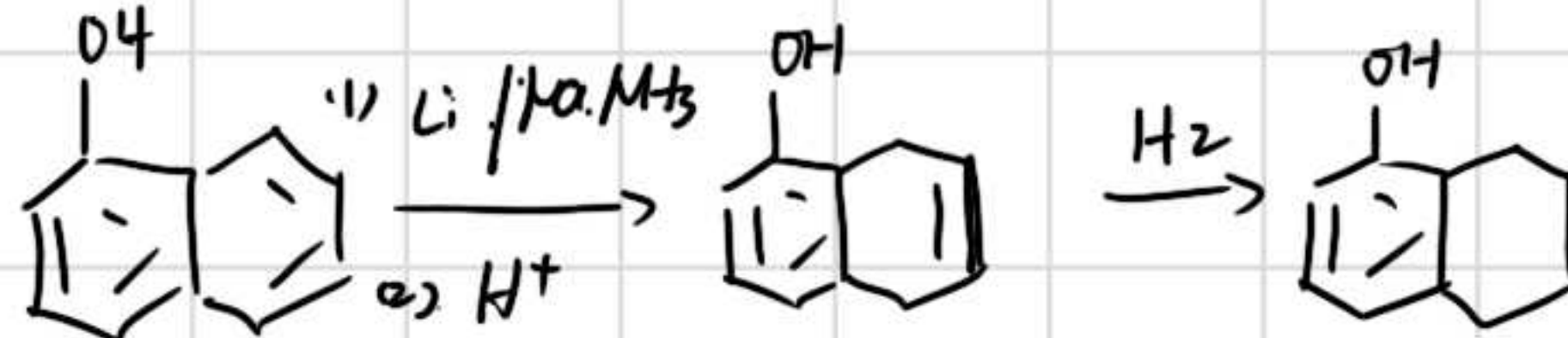
Birch 还原 (机理不要求)



因为:



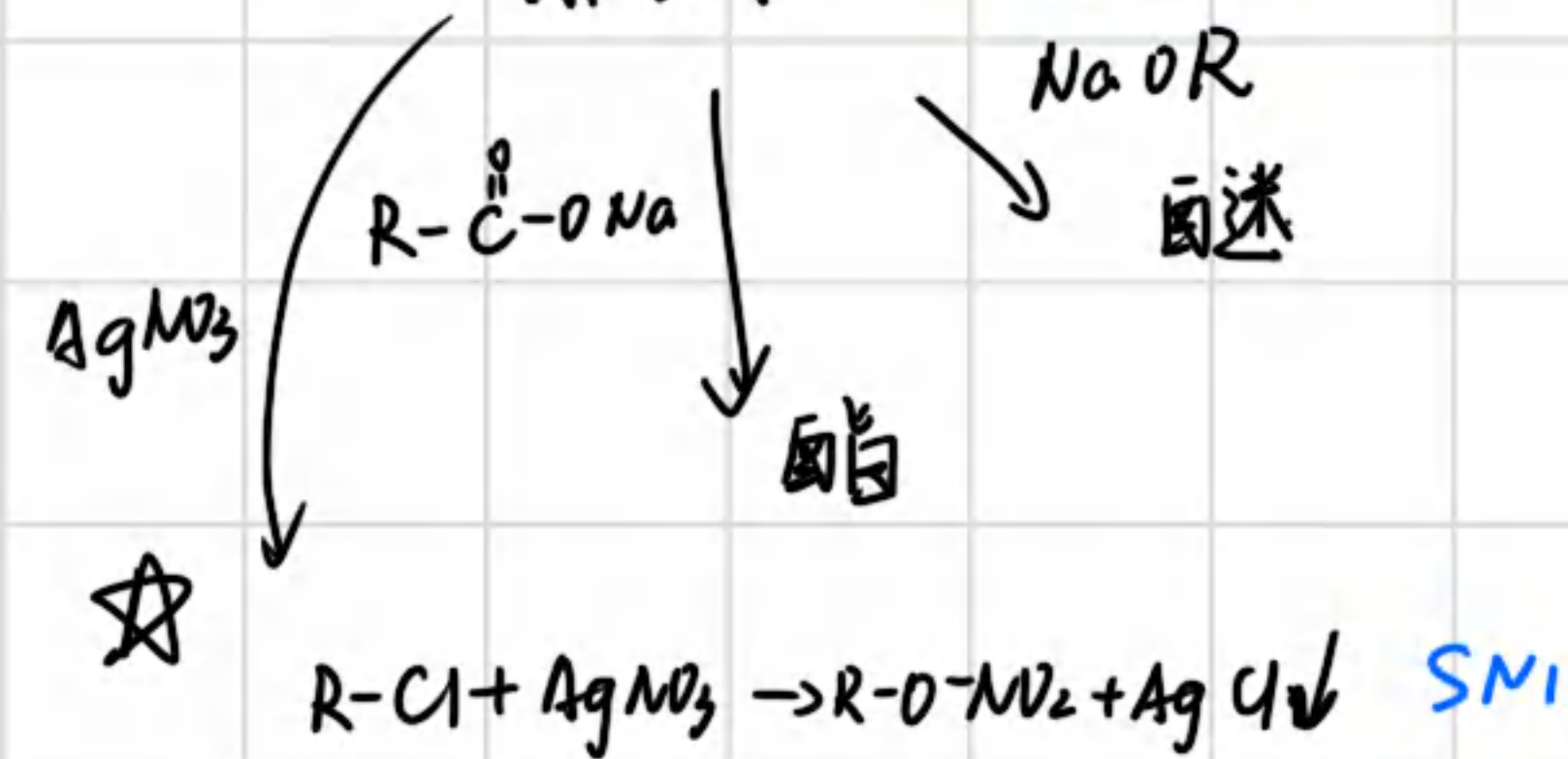
故





卤代烃  $\xrightarrow{NaCN}$  腈化物

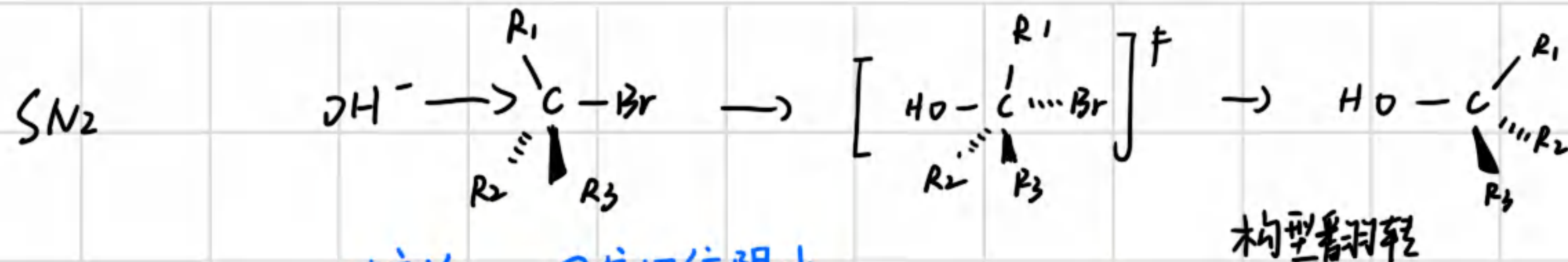
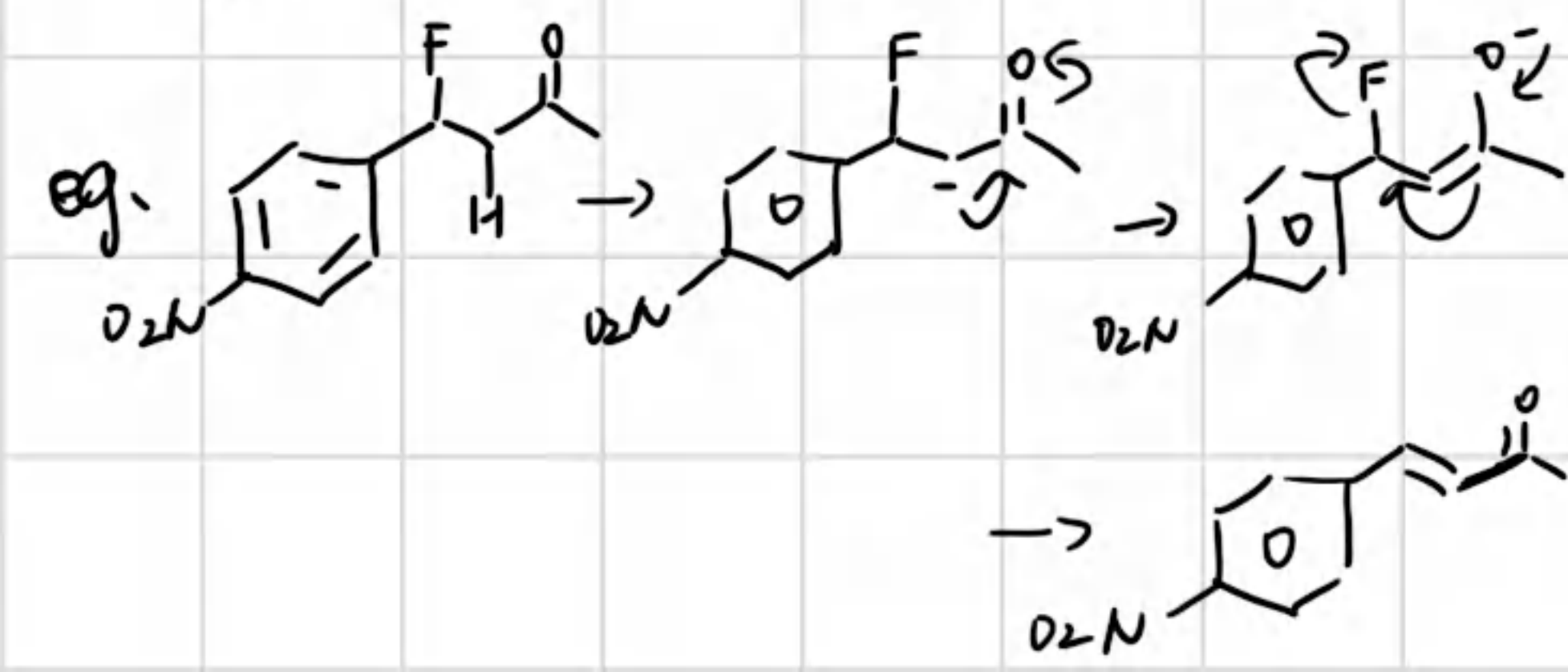
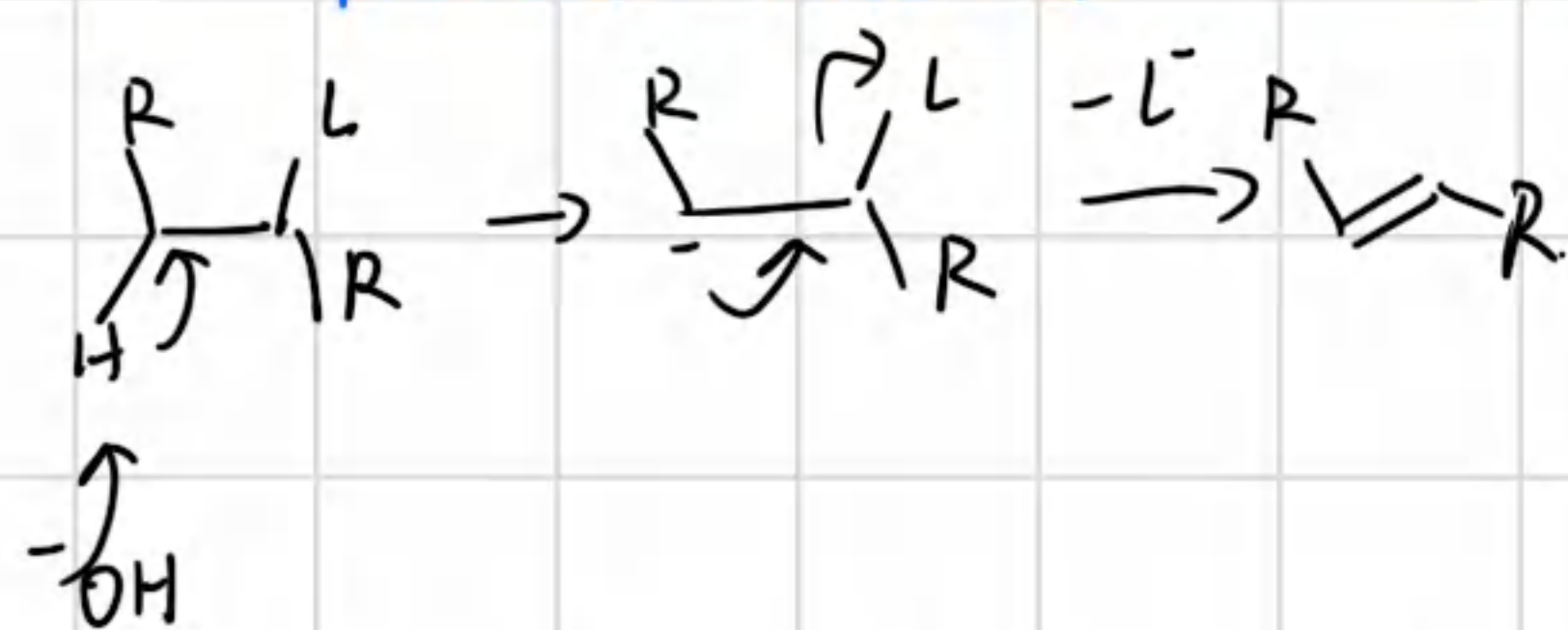
亲核取代  $\xrightarrow{NaOH \cdot H_2O}$  醇



邻基参与  $\beta$  位亲核, 邻基位于离去基反位

\*  $E1, E2$  ①  $\beta$ -H 酸性强 (连吸e<sup>-</sup>基, 如卤素, 羰基)

② 离去基难以离去 (eg. F, OH, OR)



决定性因素

① 空间位阻小

② 亲核试剂亲核力强 (反例:  $NO_3^-$ ,  $H_2O$ ,  $CH_3CH_2OH$  亲核性差, 不会  $SN2$ )

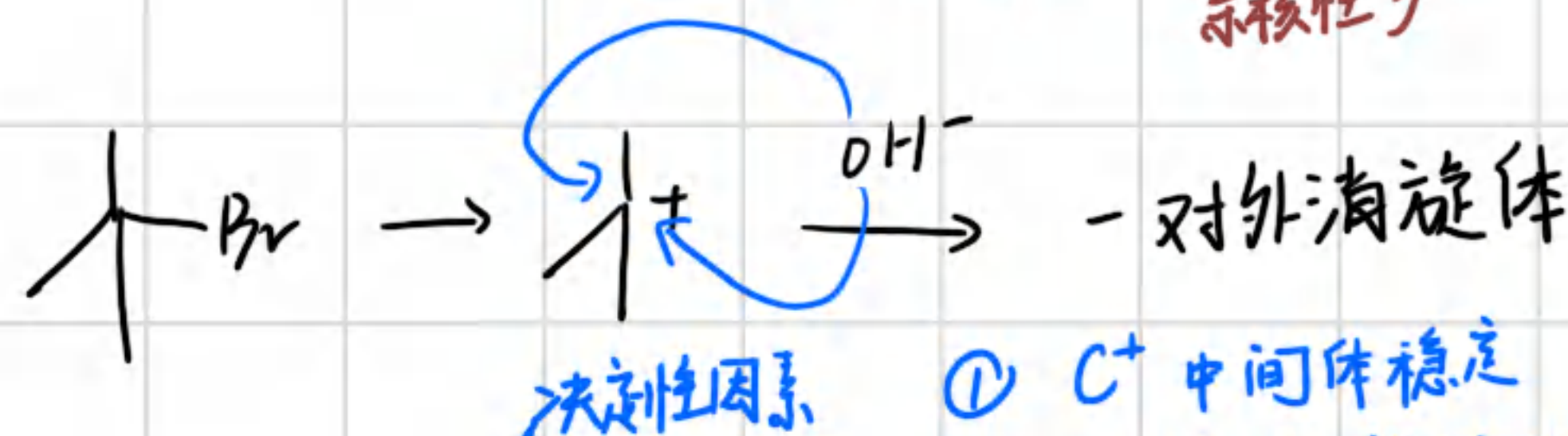
③ 极性非质子溶剂有利于  $SN2$

DMSO

丙酮, DMF

: 稳定阳离子, 使碱离子顺利解离  
亲核性  $\uparrow$

$SN1$

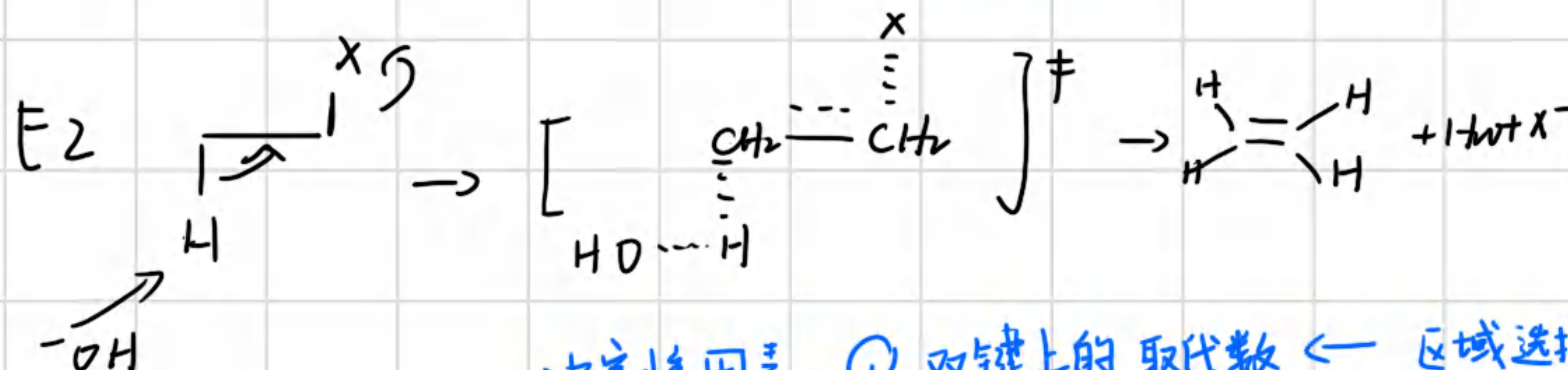


决定性因素

①  $C^+$  中间体稳定

② 允许亲核试剂亲核性强

③ 质子溶剂有利: 稳定阴离子, 使  $Br^-$  等可以稳定离去



决定性因素

① 双键上的取代数  $\leftarrow$  区域选择性

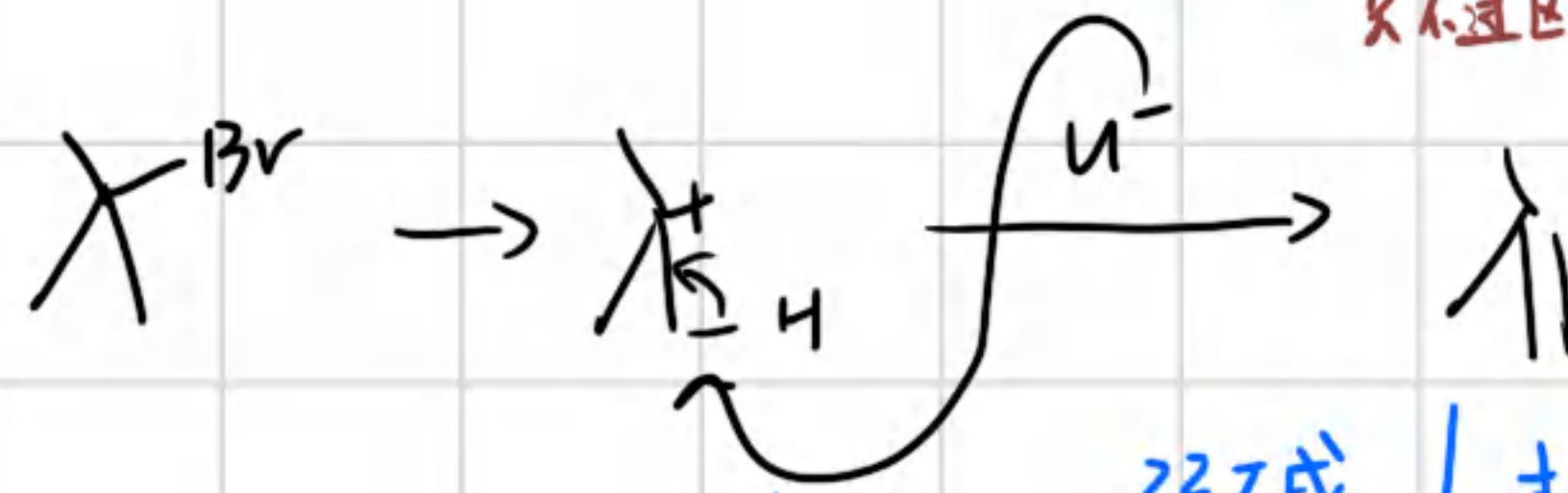
② 试剂碱性

立体选择性: 反式共面

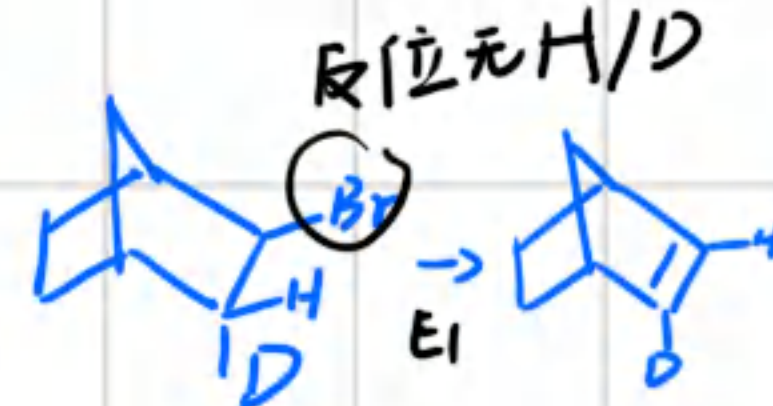
(环状: 必须位于反式直立键)

碱位阻大依旧可以  $E2$  消除  
只不过区域选择会倾向生成 Hofmann 烯烃

$E1$



弱碱 / 构象不允许反式消除





# 离去基团

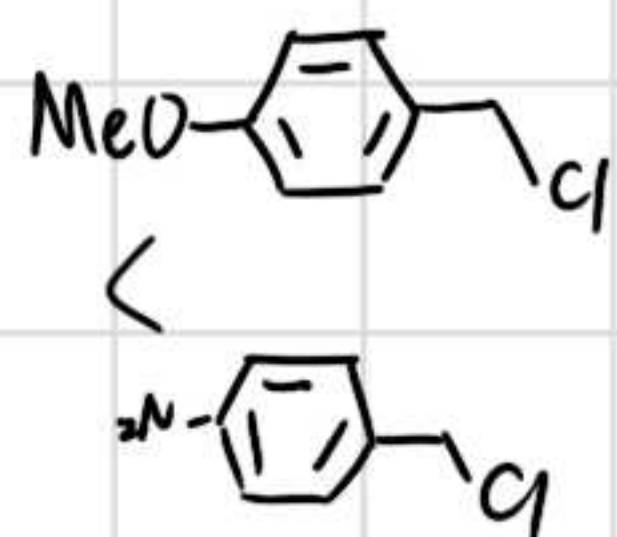
把难离去的OH变成易离去的OTs



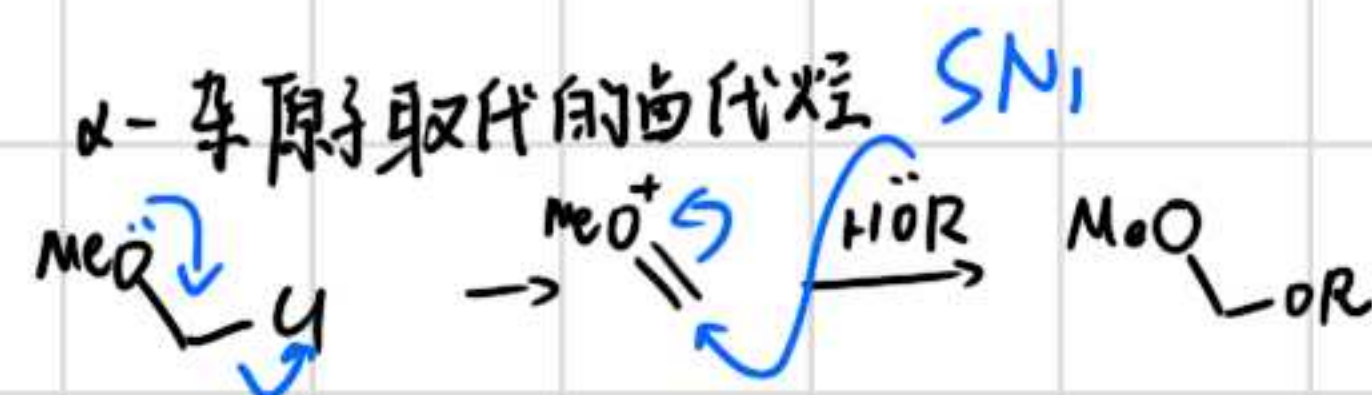
负离子稳定, 则离去能力强  $I^- > Br^- > Cl^-$

$S_N1$  or  $S_N2$

特殊:

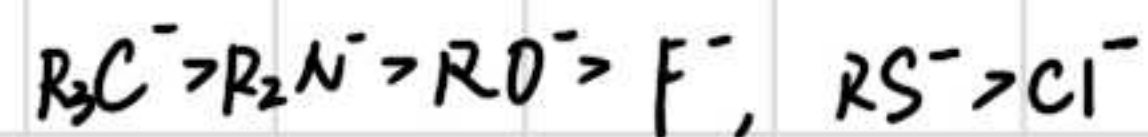


苯卤: 苯环上有给 $e^-$ 基  $\rightarrow S_N1$   
吸 $e^-$ 基  $\rightarrow S_N2$



亲核试剂

同种/同周期元素: 碱性 & 亲核性一致



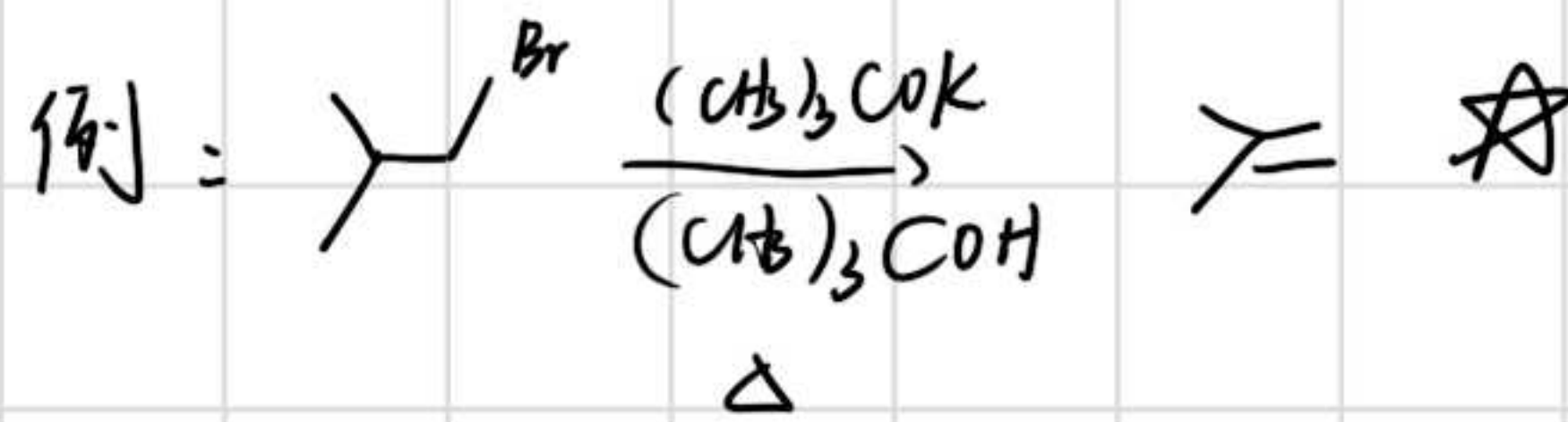
同主族: 上  $\rightarrow$  下 碱性  $\downarrow$ , 亲核性  $\uparrow$

试剂体积变大: 碱性  $\uparrow$ , 亲核  $\downarrow$



# 反应的竞争

★ 空间位阻大 / 生成更稳定产物  
倾向消除

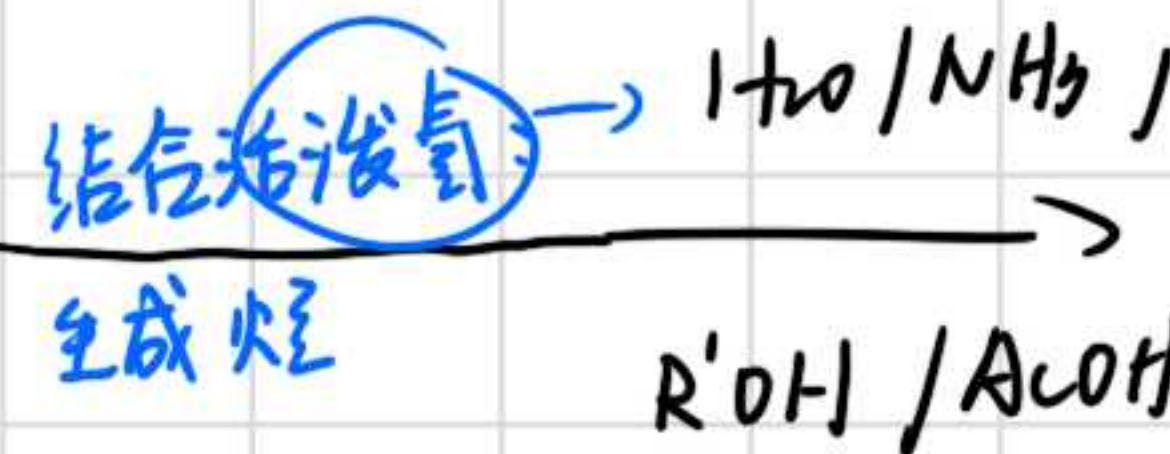


## 一般而言

|                    | 强碱 / 弱亲核                                       | 强碱 / 强亲核                                              | 弱碱 / 强亲核              | 弱碱 / 弱亲核                        |
|--------------------|------------------------------------------------|-------------------------------------------------------|-----------------------|---------------------------------|
|                    | NaH, NaNH <sub>2</sub> , LDA, BuO <sup>-</sup> | OH <sup>-</sup> , MeO <sup>-</sup> , EtO <sup>-</sup> | H <sub>2</sub> O, ROH | H <sub>2</sub> O, ROH           |
| R <sub>3</sub> CX  | E <sub>2</sub>                                 | E <sub>2</sub>                                        | S <sub>N</sub> 1      | S <sub>M</sub> / E <sub>1</sub> |
| R <sub>2</sub> CHX | E <sub>2</sub>                                 | E <sub>2</sub>                                        | S <sub>N</sub> 2      |                                 |
| RCH <sub>2</sub> X | E <sub>2</sub>                                 | E <sub>2</sub> / S <sub>N</sub> 2                     | S <sub>N</sub> 2      |                                 |
| CH <sub>3</sub> X  | S <sub>N</sub> 2                               | S <sub>N</sub> 2                                      | S <sub>N</sub> 2      |                                 |

## 卤代烃与金属

格氏试剂

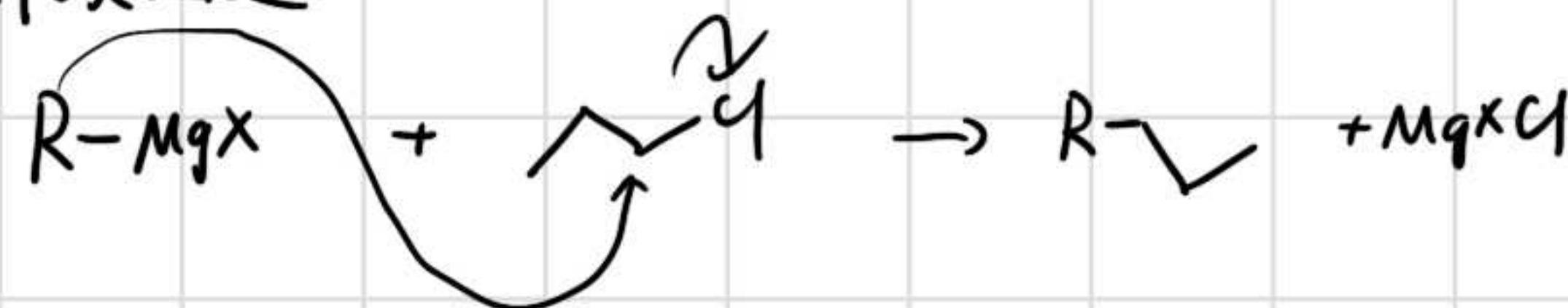


制备格氏试剂:



作为强亲核试剂去进攻

eg. 作碳链延长





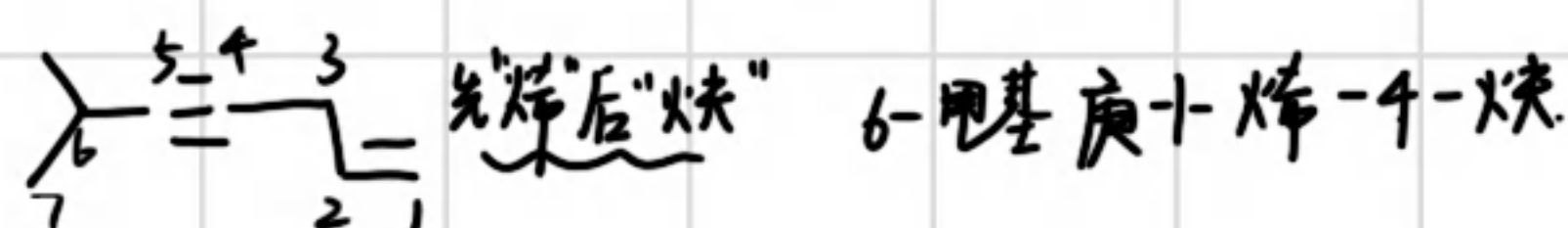
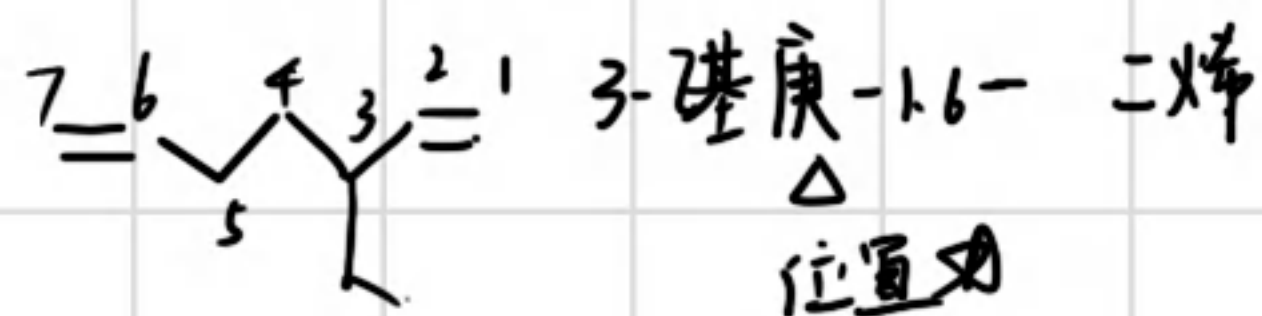
命名 -iso 异 甲基: Me  
-neo 新 乙基: Et

选主链: 碳链长  $\rightarrow$  重键多  $\rightarrow$  重键位次组低  $\rightarrow$  取代基最多  $\rightarrow$  取代基位次组最低

编号: 重键最低  $\rightarrow$  取代基最低  $\rightarrow$  取代基英文字母在前  
(先比第一个, 再比第二个...)

-iso 先于 -neo

-Et 先于 Me



## 烯烃构型

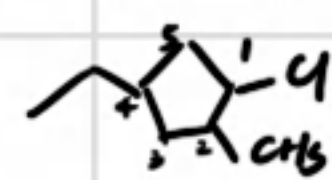
①. 顺. 反

(cis) (trans)

② 优先基团 同侧 Z 每一层都先比最大的  
异侧 E

环的命名: 链原子数  $>$  环  $\rightarrow$  链为母体  
环为取代基

编号遵循最低位次组原则, 且英文字母靠前的与前面

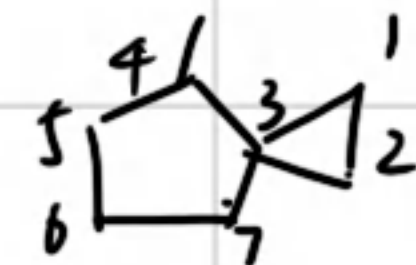


1-氯-4-乙基-3-甲基环戊烷

4-甲基螺[2.4]庚烷

7-甲基螺[4.4]壬-2-烯

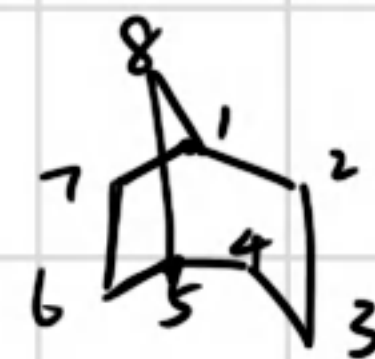
螺环



编号: 小环

$\rightarrow$  大环

桥环 桥: 长  $\rightarrow$  短



双环-[3.2.1]辛烷



立体

直立: a 键

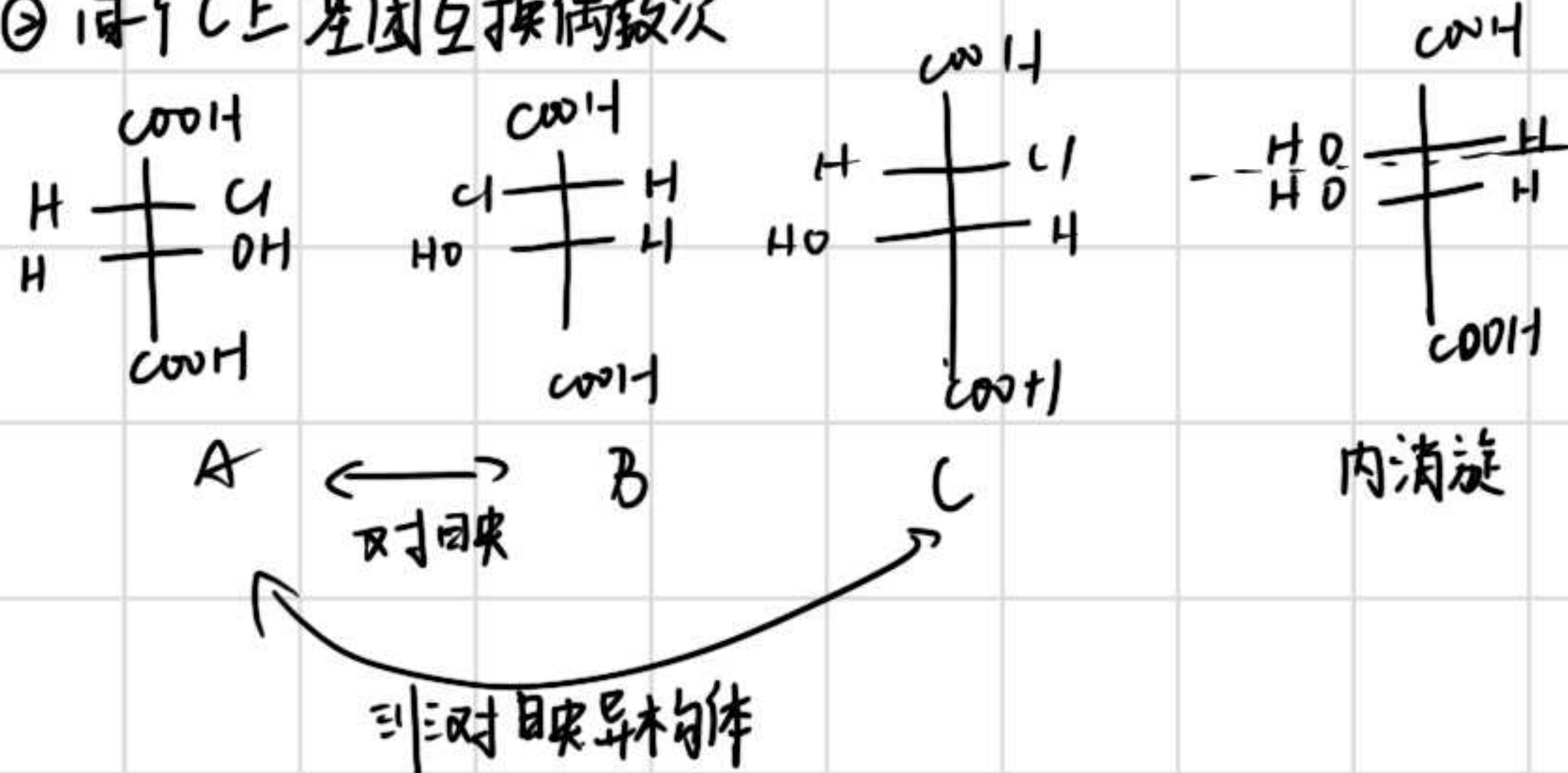
大体积基团: 1,3 直立键相互作用

平伏: e 键

Fisher 投影式: 横键向前, 竖键向后  
相同:

① 转 180

② 同一 C 上基团互换偶数次





# 定位效应

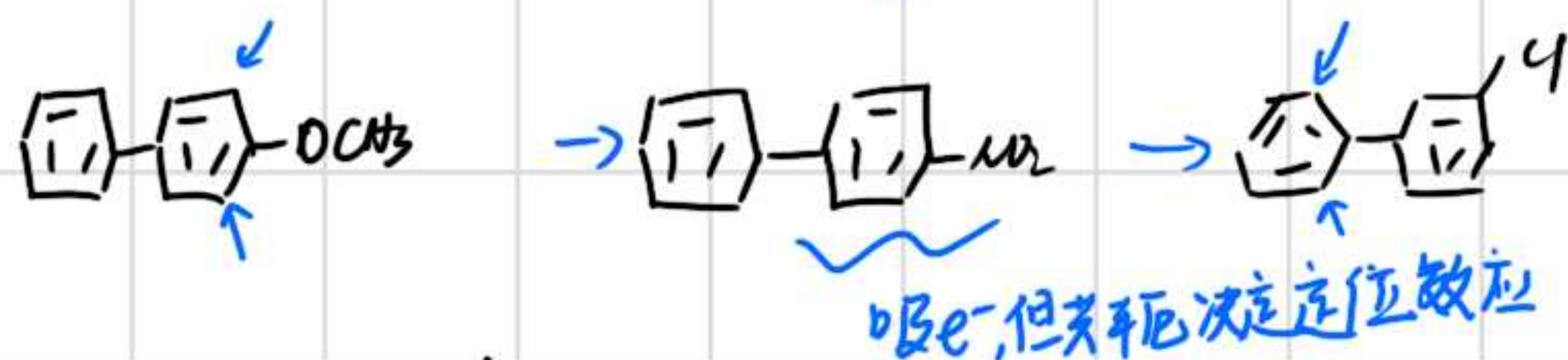
① Z取代基

- 定位效应一致 → 位阻小
- 定位效应不一致 → 同为吸/给 由强的决定
- 吸+给 由“给”决定

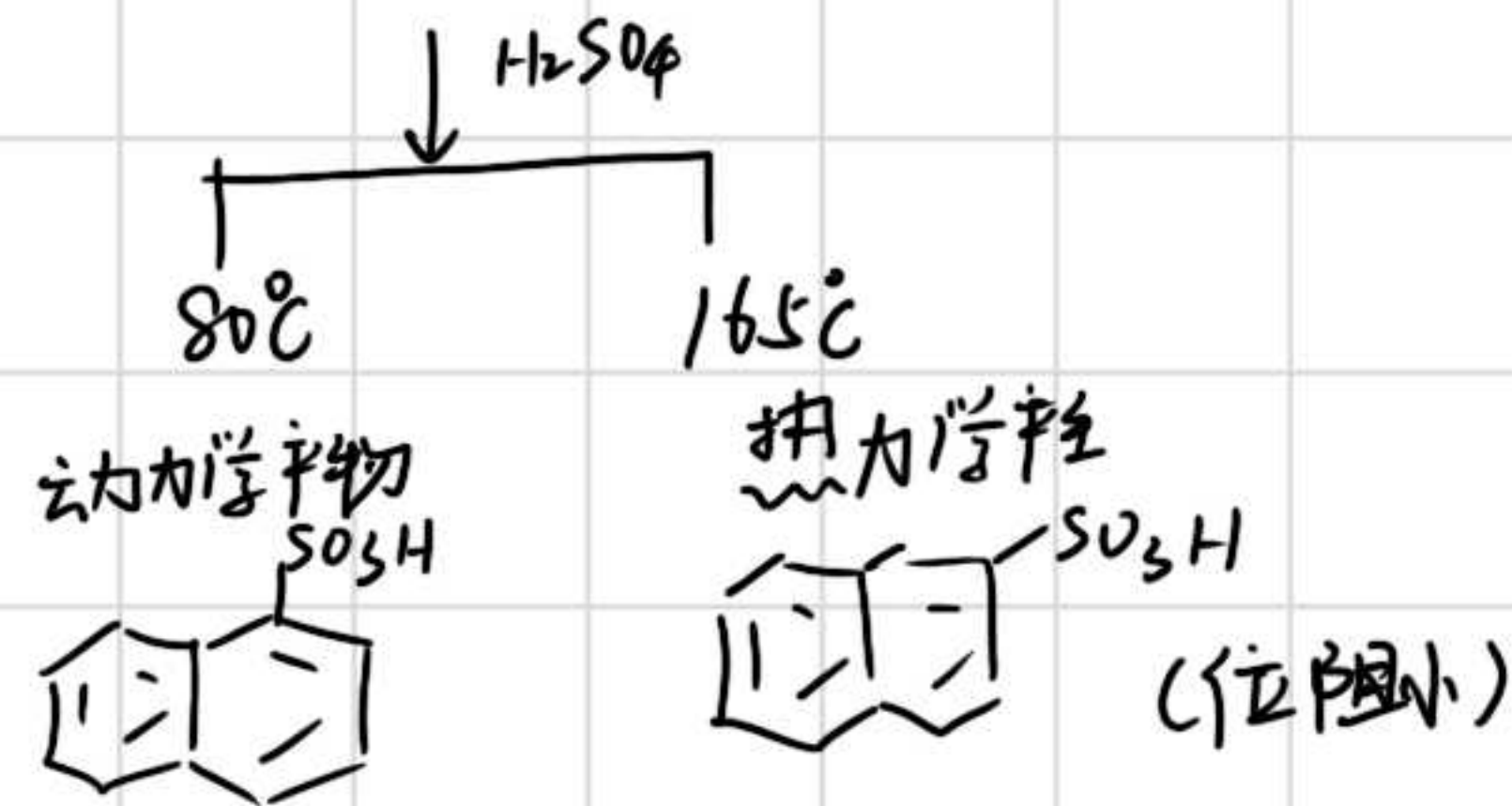
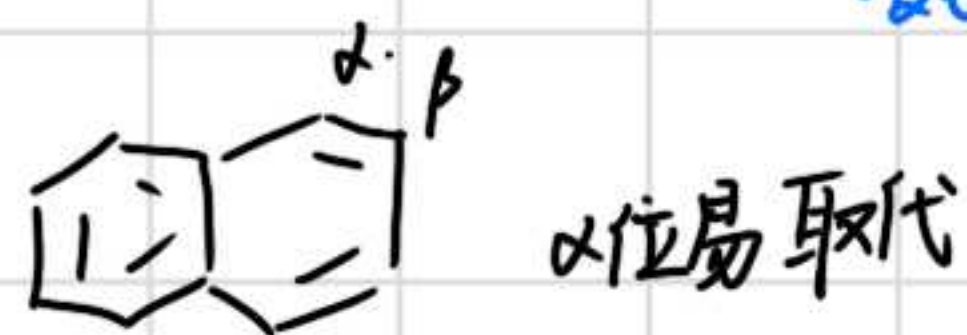
## ② 多环芳烃



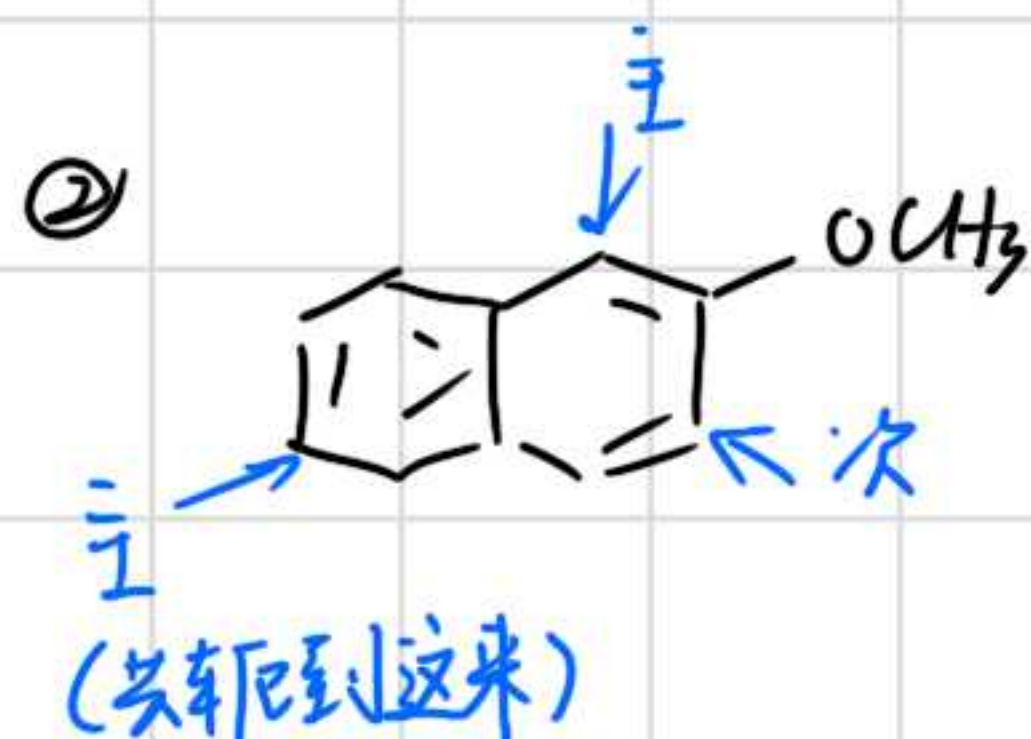
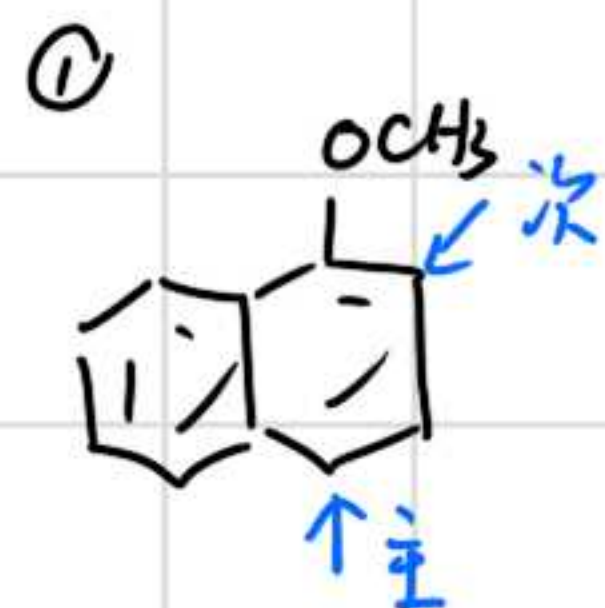
视为取代基, 活化A的邻对位, 对位的位阻小, 更优



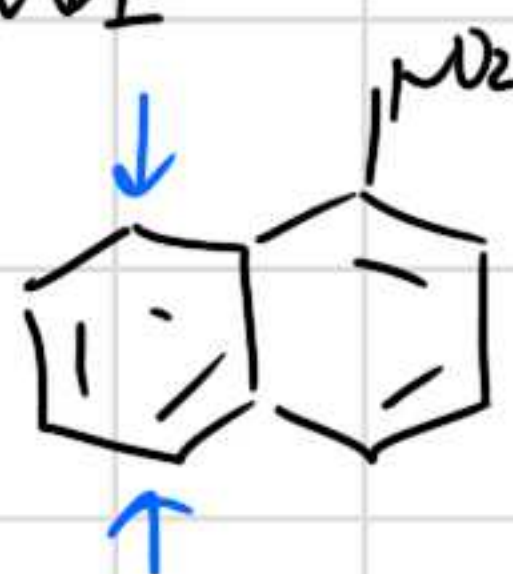
## ③ 萘



## 给电子基:



## 吸电子基



## 邻对位活化

-NHCO<sub>2</sub>R

-OR

-Ph

## 邻对位钝化

-NO<sub>2</sub>

-Cl

## 间位

CO<sub>2</sub>R, C=O, NH<sub>2</sub>...



# 醇亲核取代

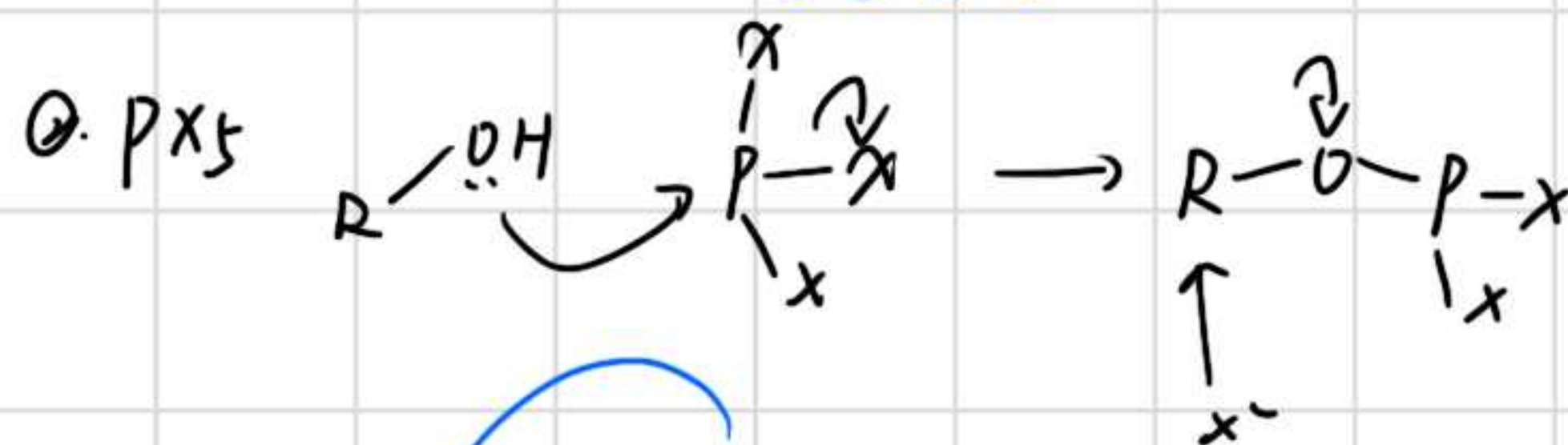
①.  $HX$   $HI > HBr > HCl$

1° ROH:  $S_N2$  /  $S_N1$   
2° ROH, 3° ROH:  $S_N1$

下/烯 > 3级 > 2级 > 1级

☆ Lucas 试剂:  $HCl / ZnCl_2$  与醇反应生成混浊 (不溶的卤代烷)

芳, 烯丙, 叔, 仲快  
伯: 10min后  
伯: 需加热



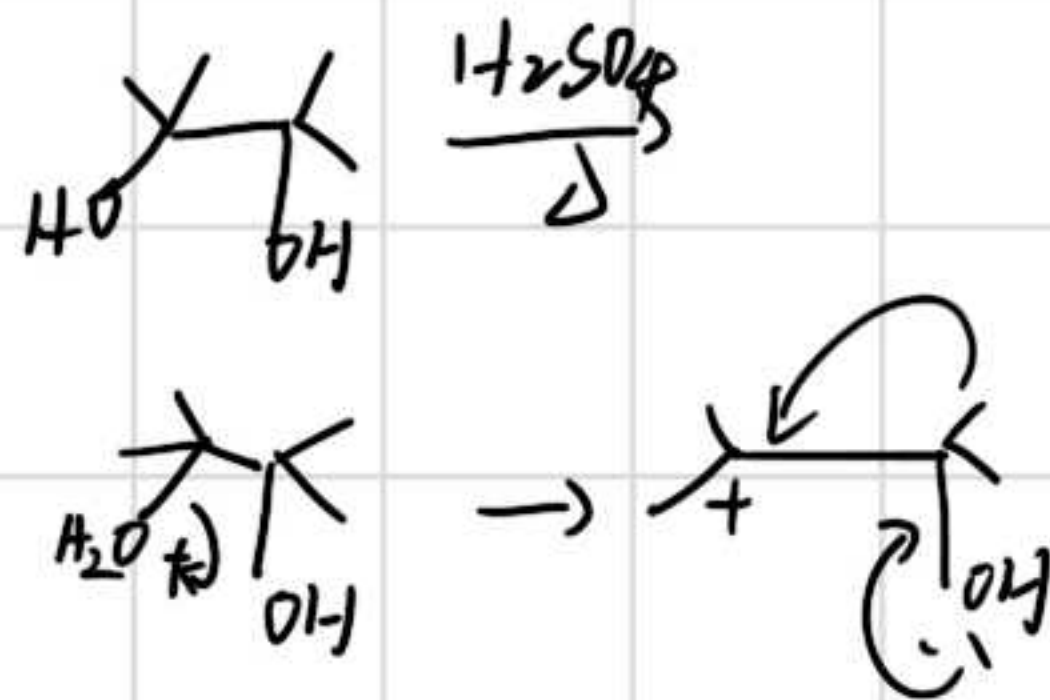
构型保持

③  $SOCl_2$

构型保持

~ 翻转

频哪醇重排



迁移: 迁移基团与离去基团反式

区域: 更稳定  $C^+$  优先产生

迁移:  $\text{环己基} > H > CR_3 > HCR_2$

(富e-基团优先)

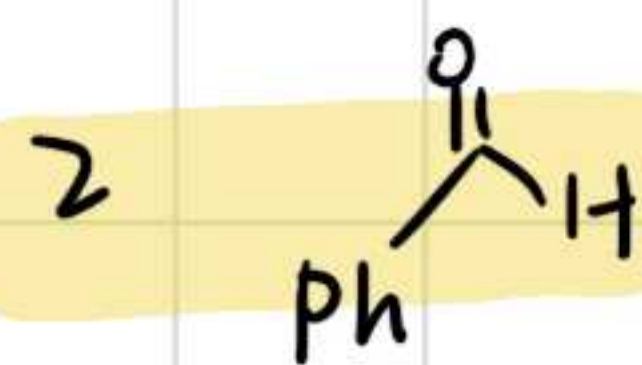
环氧开环

酸 进攻  $C^+$  稳定的C  
碱 进攻位阻小的C

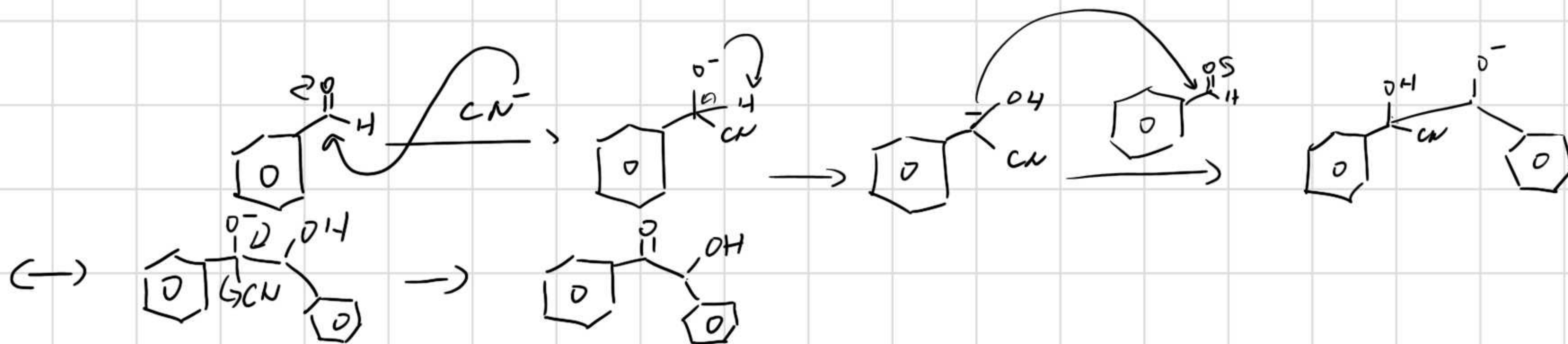
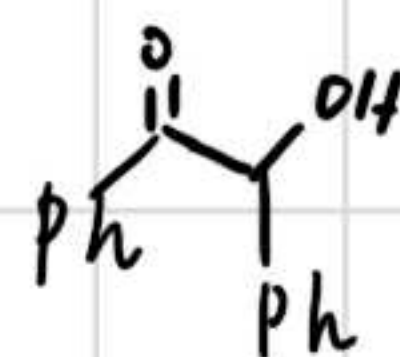


醛

安息香缩合



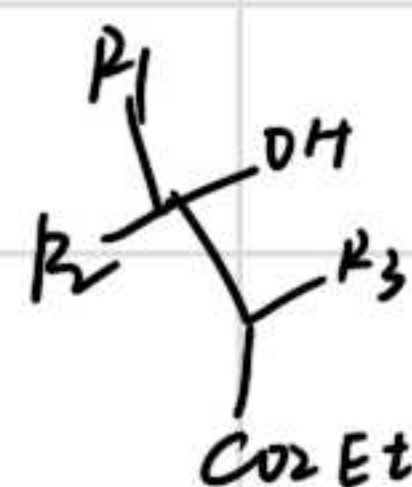
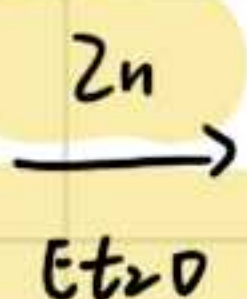
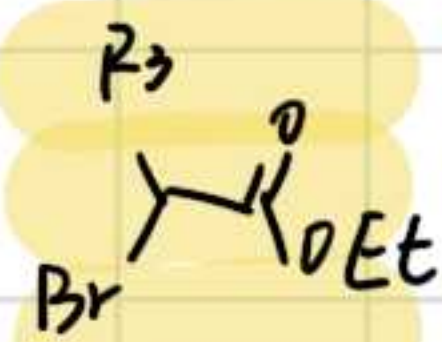
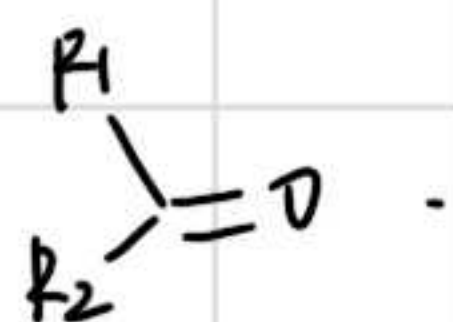
$\xrightarrow{\text{CN}^- (\text{催化量})}$



Reformatsky (有机锌试剂)

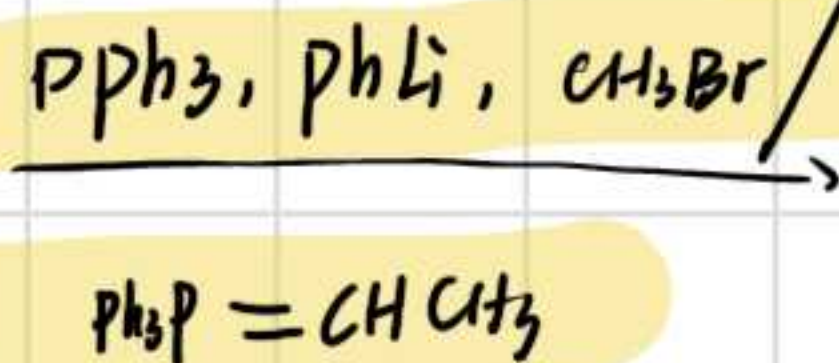
↓ R与醛酮同反应

相对于格氏试剂, 选择性优





# Wittig

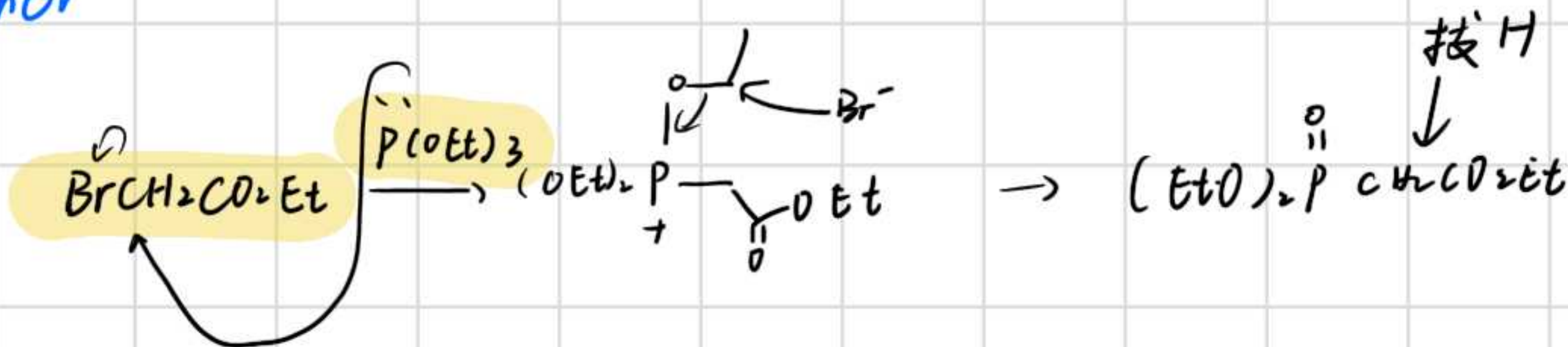


制备环外双键



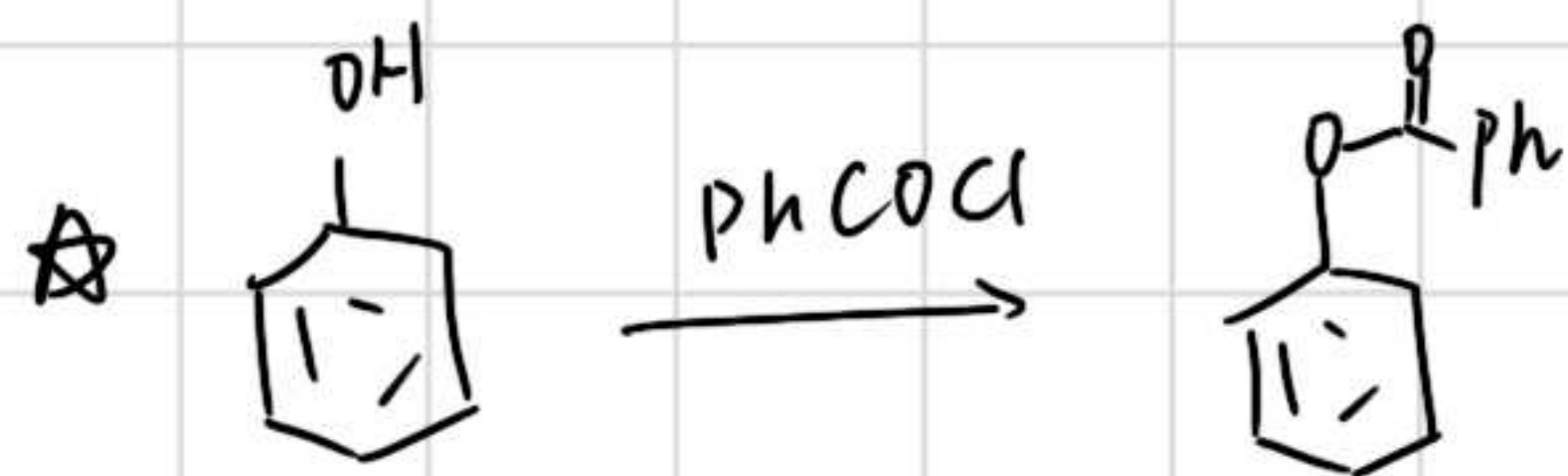
四元环机理  $\rightarrow$  反式双键！

## Wittig-Hornor



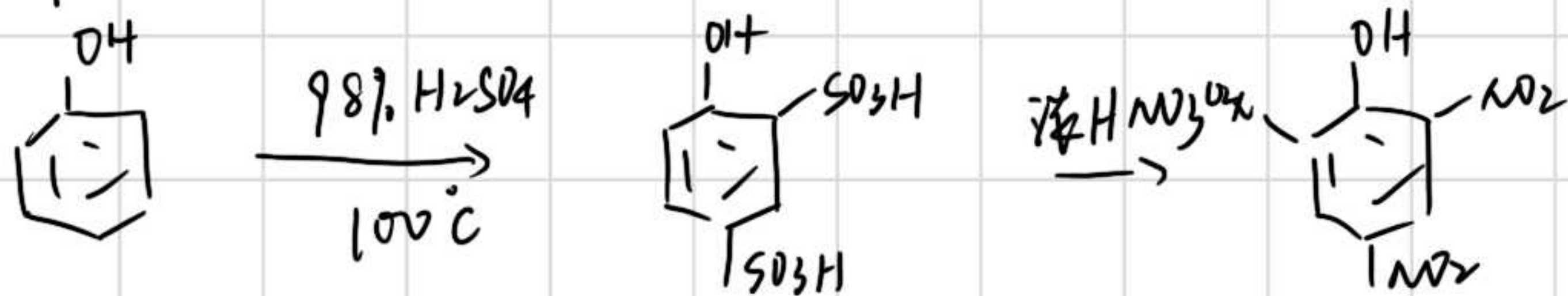
合成  $\alpha, \beta$  不饱和酯

## 酚

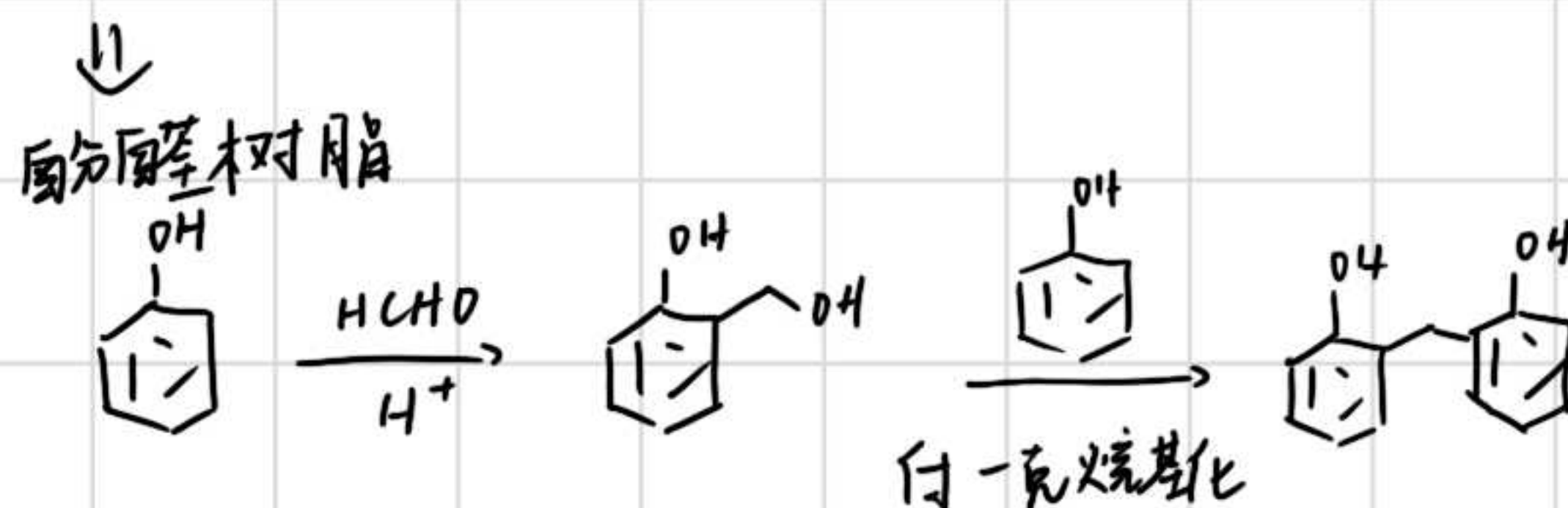


2 酚  $\rightarrow$  酯

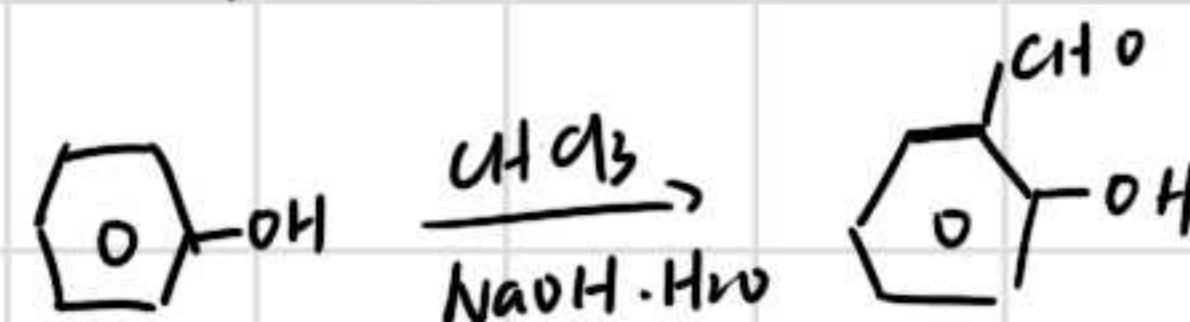
★ 苯酚易被  $HNO_3$  氧化，但不会被浓  $H_2SO_4$  氧化



付-克烷基化：一般对位



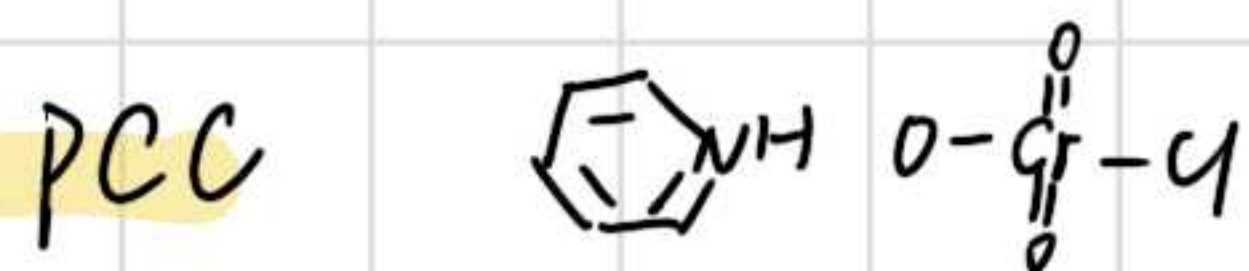
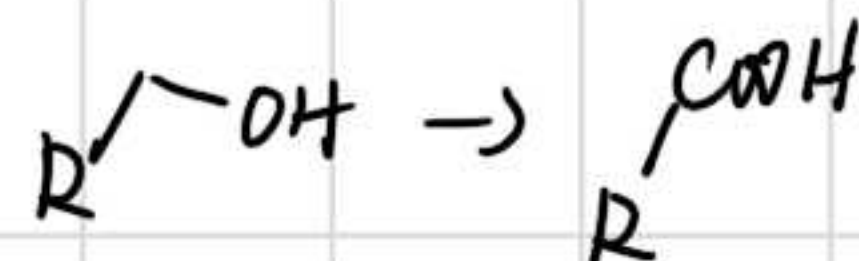
Reimer-Tiemann



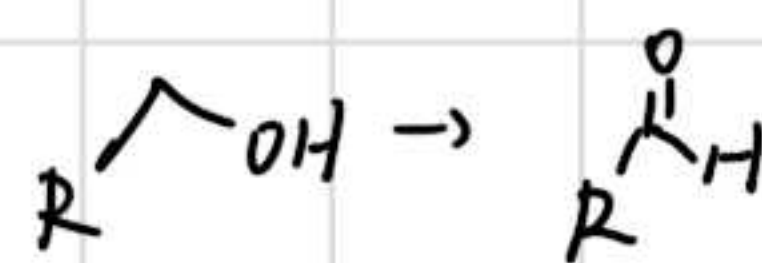


# 氧化剂

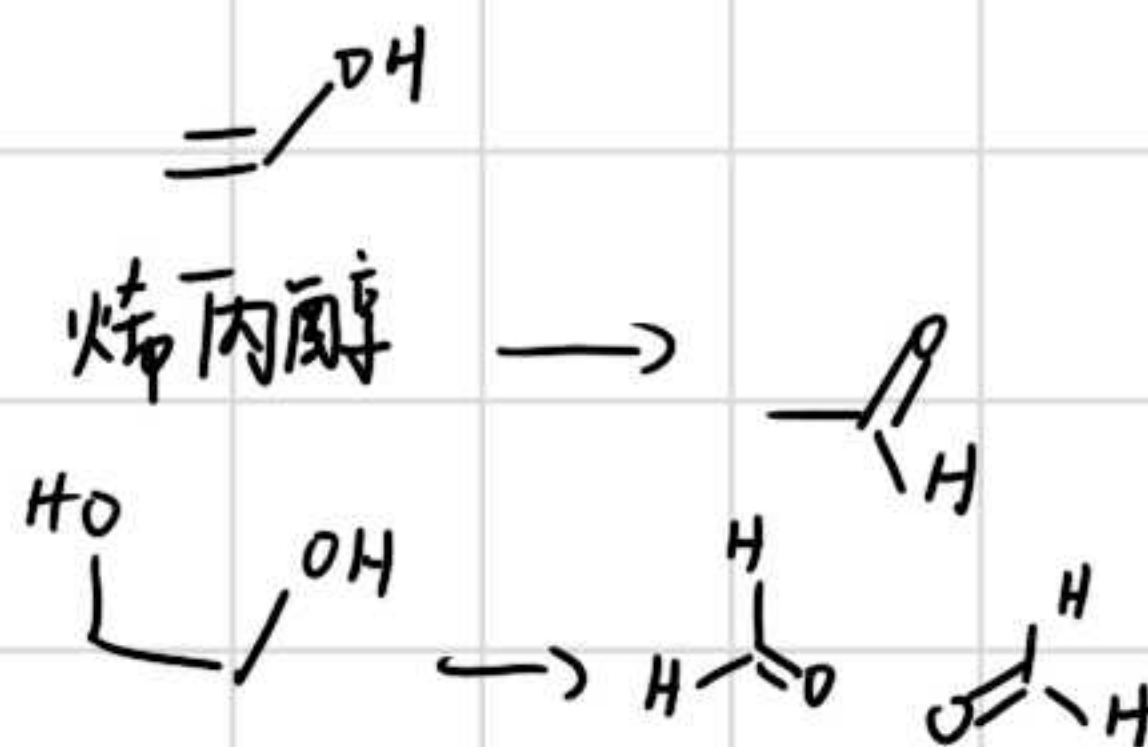
Johns ( $\text{CrO}_3, \text{H}_2\text{SO}_4$ )



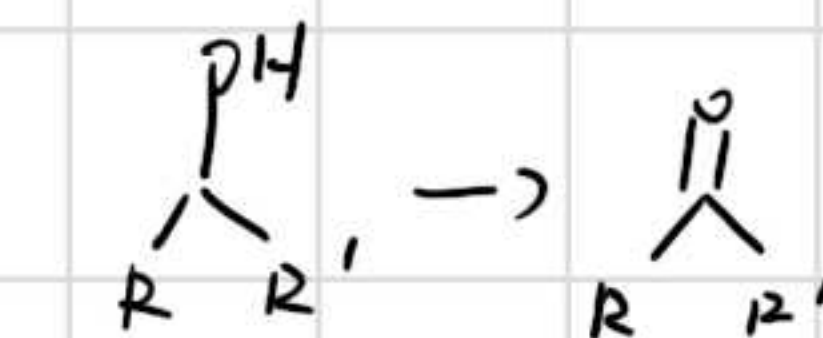
Sarrette, Collins



$\text{MnO}_2$

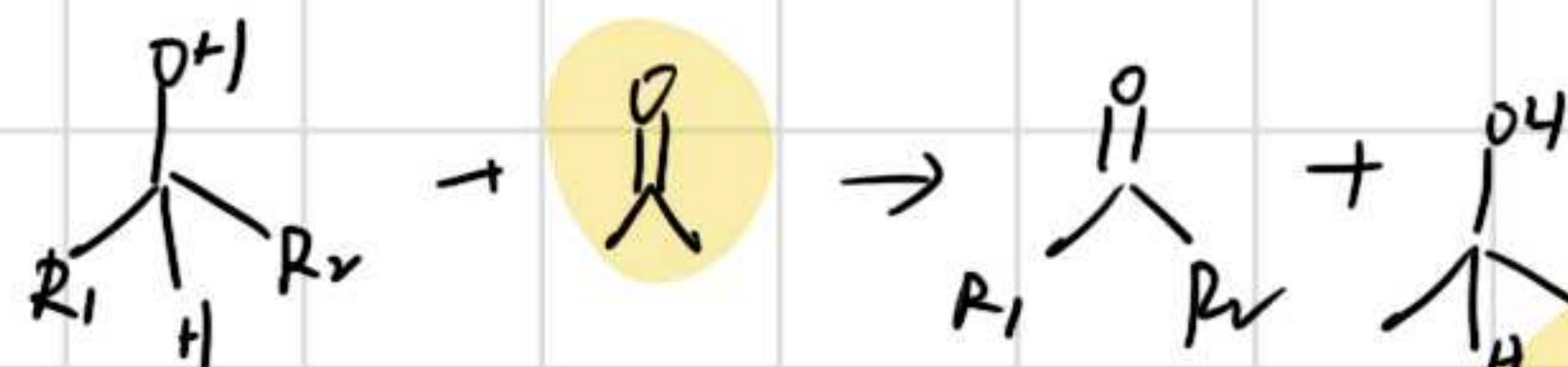


Swern  $\text{DMSO}$   $\text{NEt}_3$   $(\text{COCl})_2$



Oppenauer  $\text{Al}(\text{O}i\text{Pr})_3$

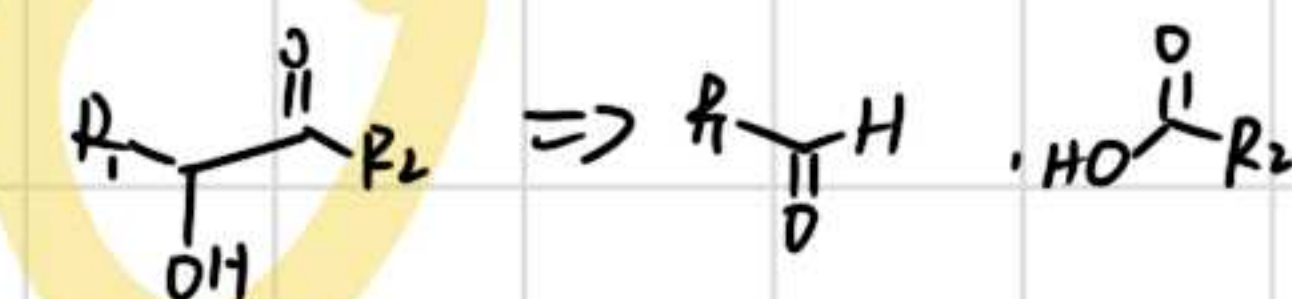
氧化不饱和醇 / 含有对酸不稳定基团的仲醇



$\text{Pb}(\text{OAc})_4$   $\text{HIO}_4$

邻二醇断 C-C  $\Rightarrow$  醛 / 酮

立体: 顺 / 能形成双平伏反



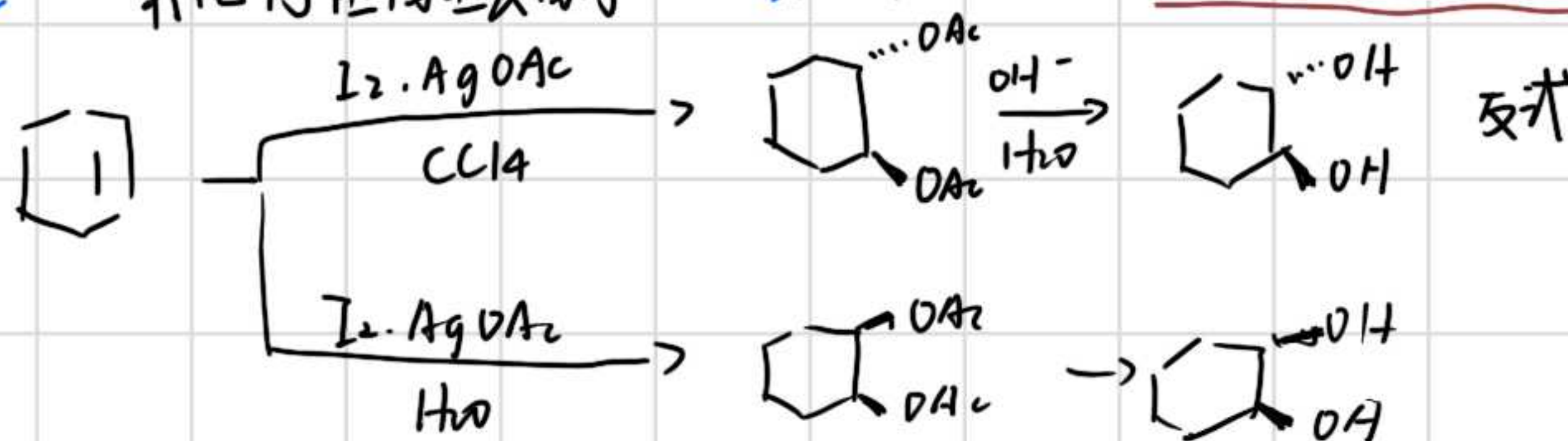
$\text{OsO}_4$  烯  $\rightarrow$  顺式邻二醇

$\text{O}_3$  +  $\text{Zn} / \text{SMe}_2$

氧化停在醛及酮

烯炔 +  $\text{O}_3$ : 双键反应, 三键不反应  $\leftarrow$  "选择性"

$\text{I}_2, \text{AgOAc}$





$KMnO_4$

炔 烯  
— 全部为酸 /  $CO_2$  / 同同  
— 碱性 — 邻二醇  
— 酸性

Wacker 烯  $\xrightarrow{H_2O, Cu^+, Pd^{2+}}$  同同 / 同同

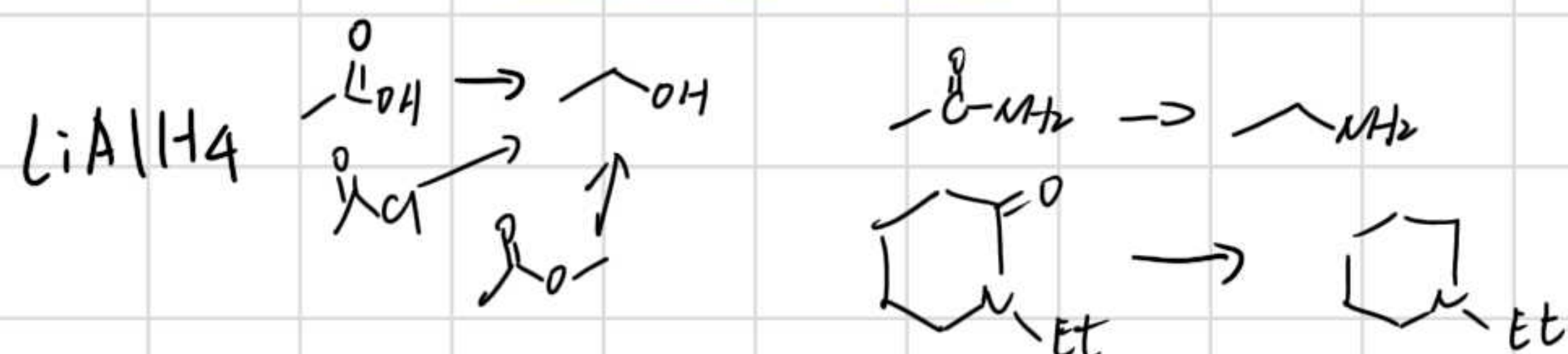
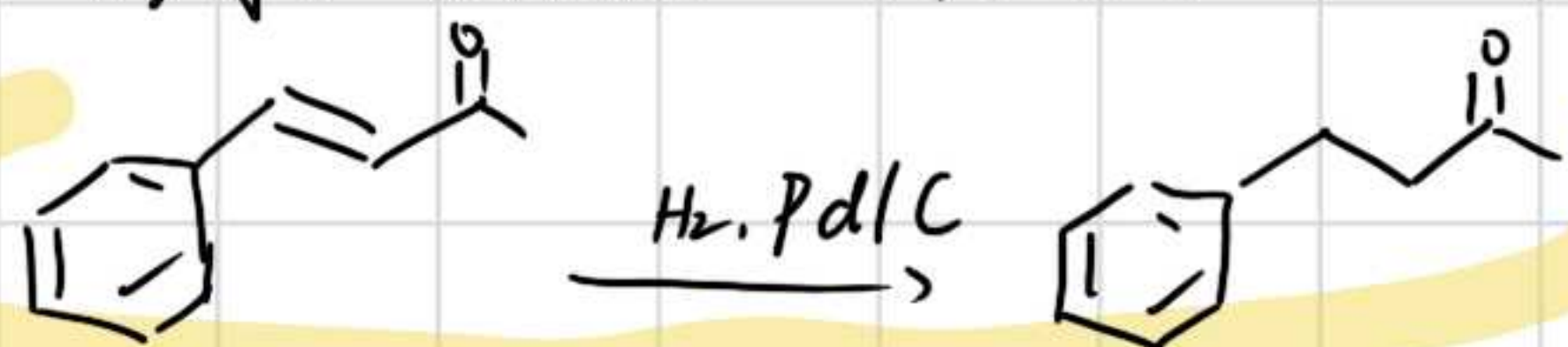


# 还原

Lindlar Pd : ( $\text{CaCO}_3/\text{Pd}$ ,  $\text{PdO}$ ,  $\text{Pd}(\text{OAc})_2$ ,  $\text{Pd/BaSO}_4$ ) 炔  $\rightarrow$  顺式烯烃

$\text{Na/Li}$ ,  $\text{NH}_3(\text{liq})$  炔  $\rightarrow$  反式烯烃

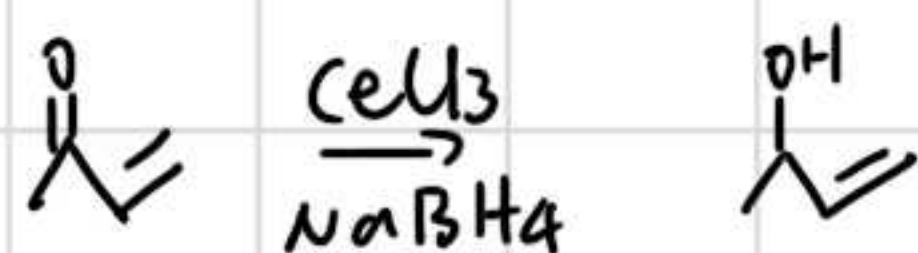
$\text{H}_2$ ,  $\text{Pd/C}$  : 只与双键反应, 不与羰基反应



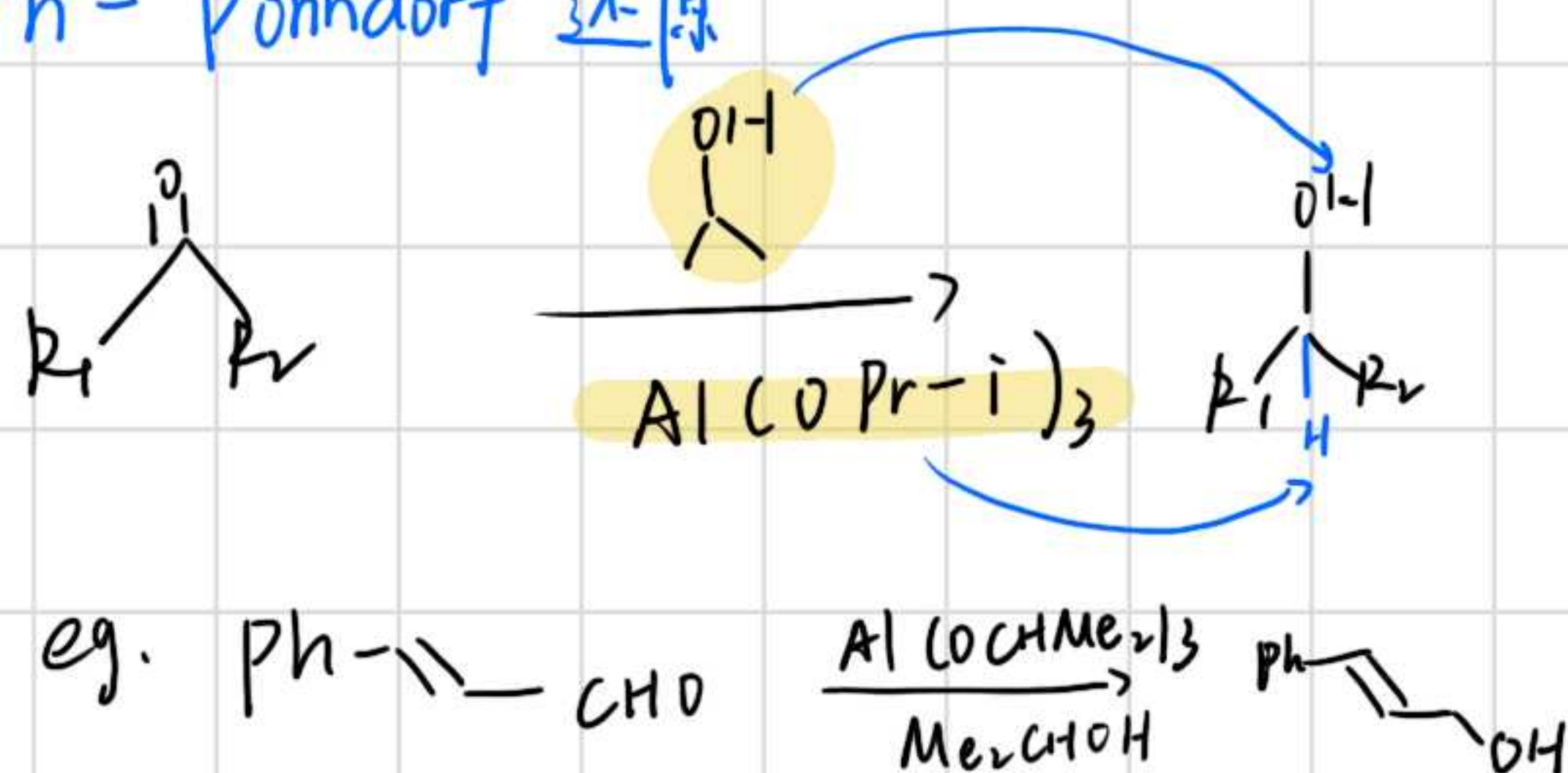
酰氯  $\rightarrow$  醛  
Rosenmund 还原



$\text{NaBH}_4$  : 亚胺, 醛, 酮, 与它们共轭的双键

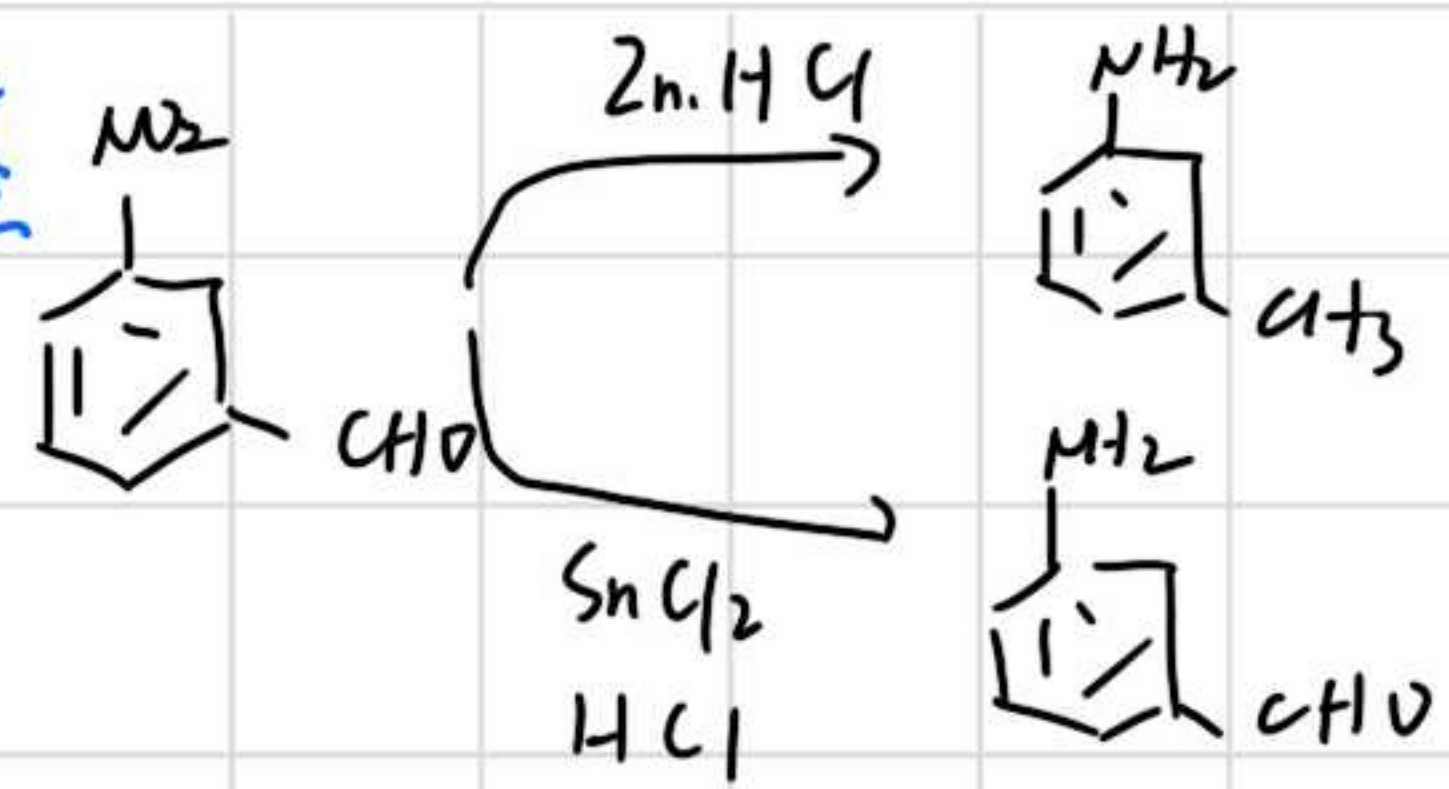


Meerwein-Ponndorf 还原

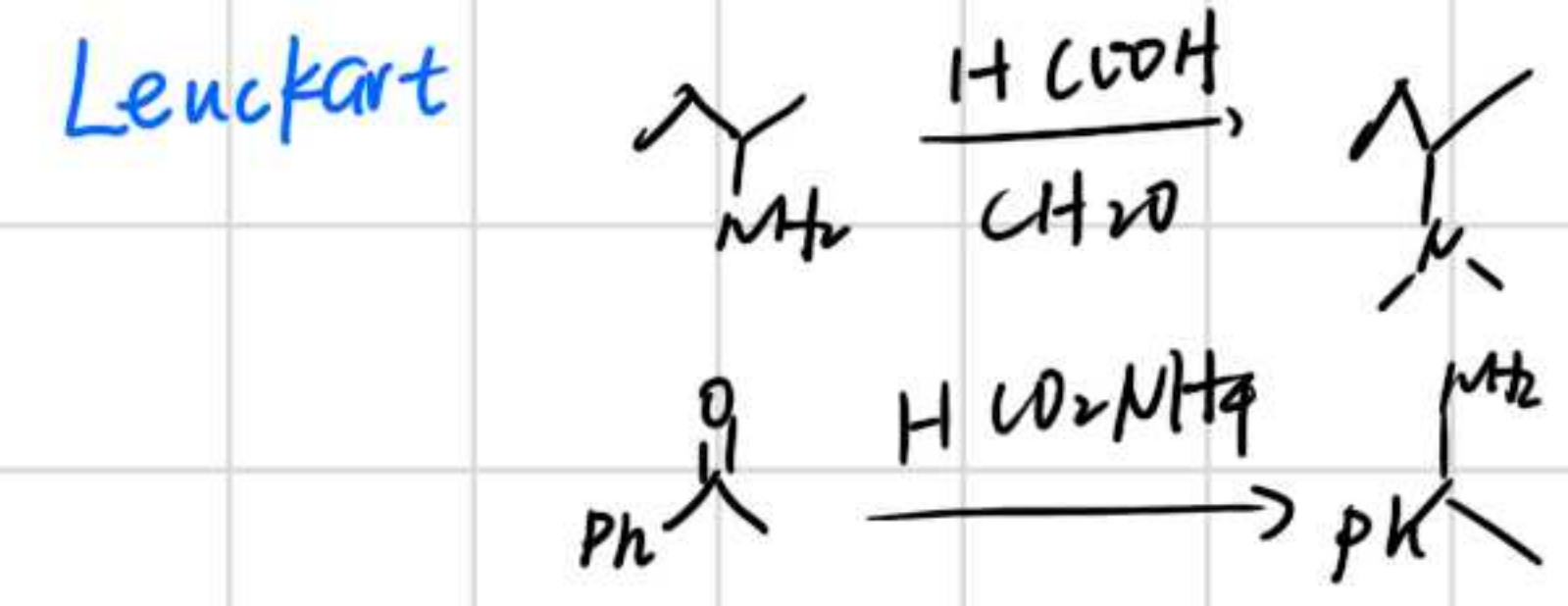
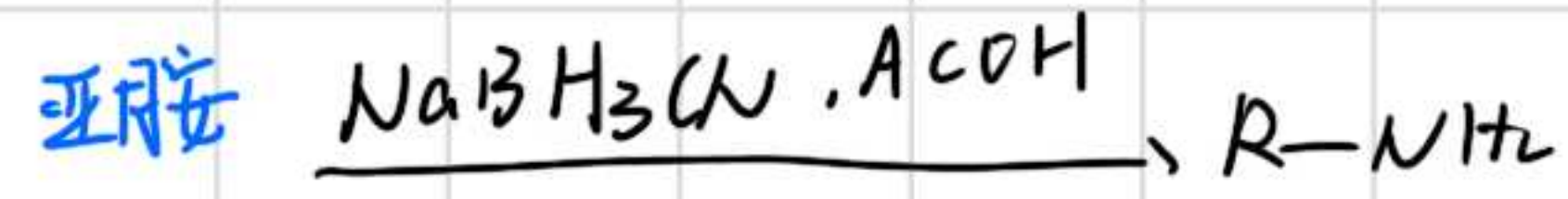




硝基

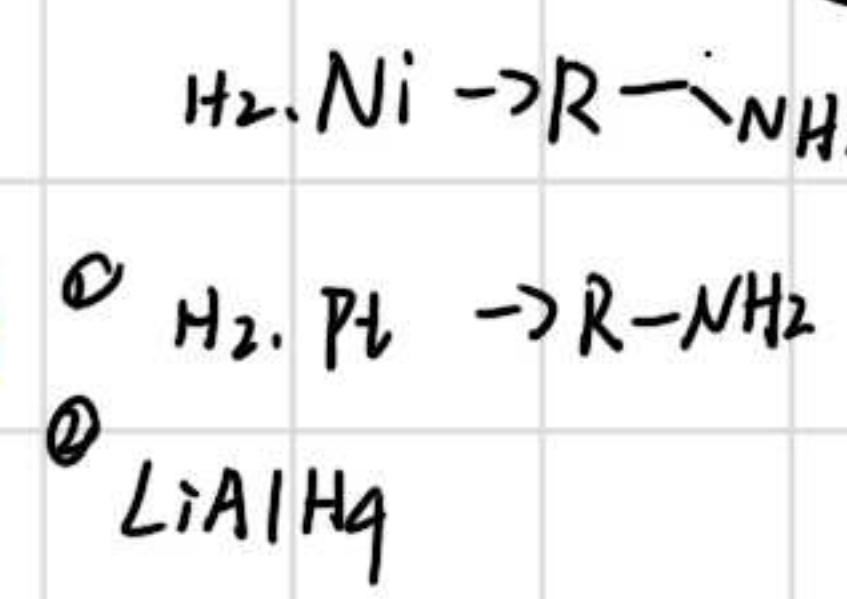


肟



肟

重氮



芳香性

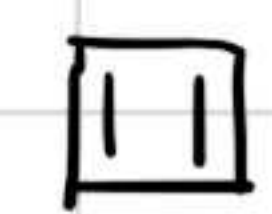
反芳香性

非芳香性

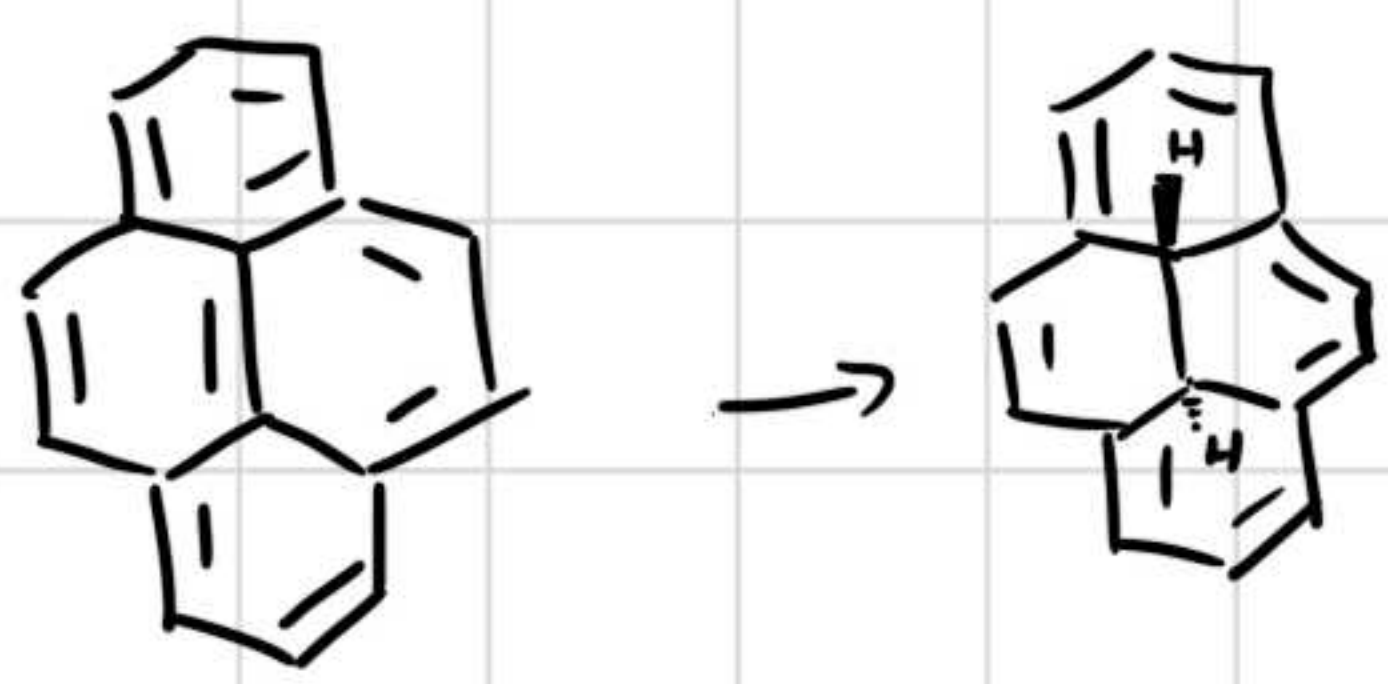
平面, 单双交替, 环闭,  $(4n+2)$  个  $e^-$



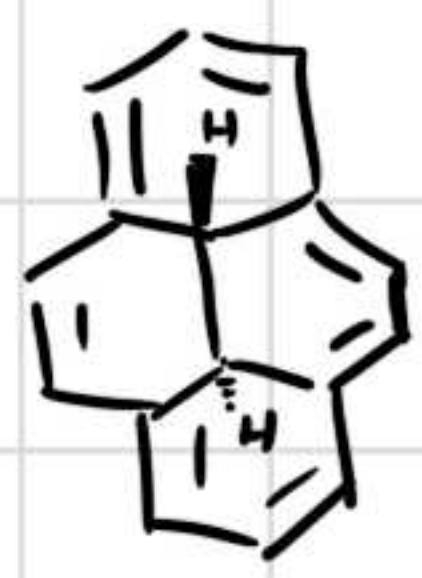
平面, 单双交替, 环闭,  $4n$  个  $e^-$



单双交替 不共面

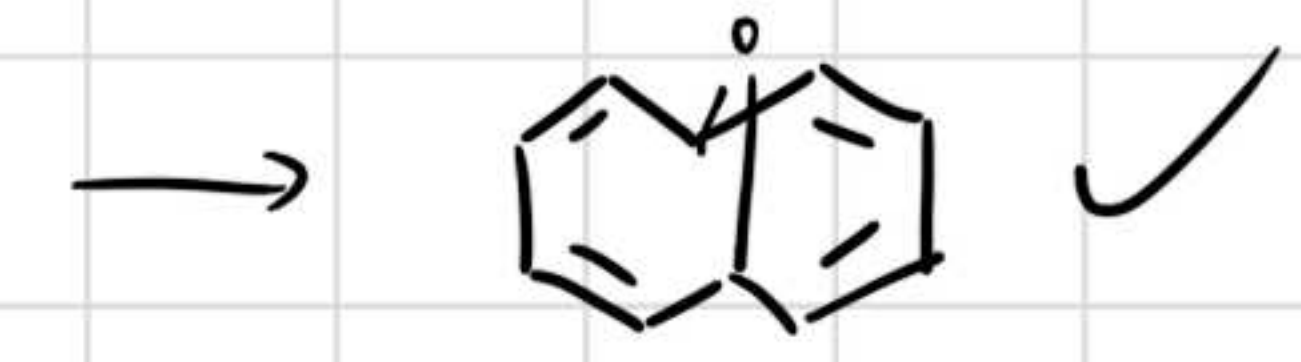


✓  
外圈 14 个  $e^-$



✓

[10]-轮烯: 不共面



[14]-轮烯; [8]-轮烯 ✓



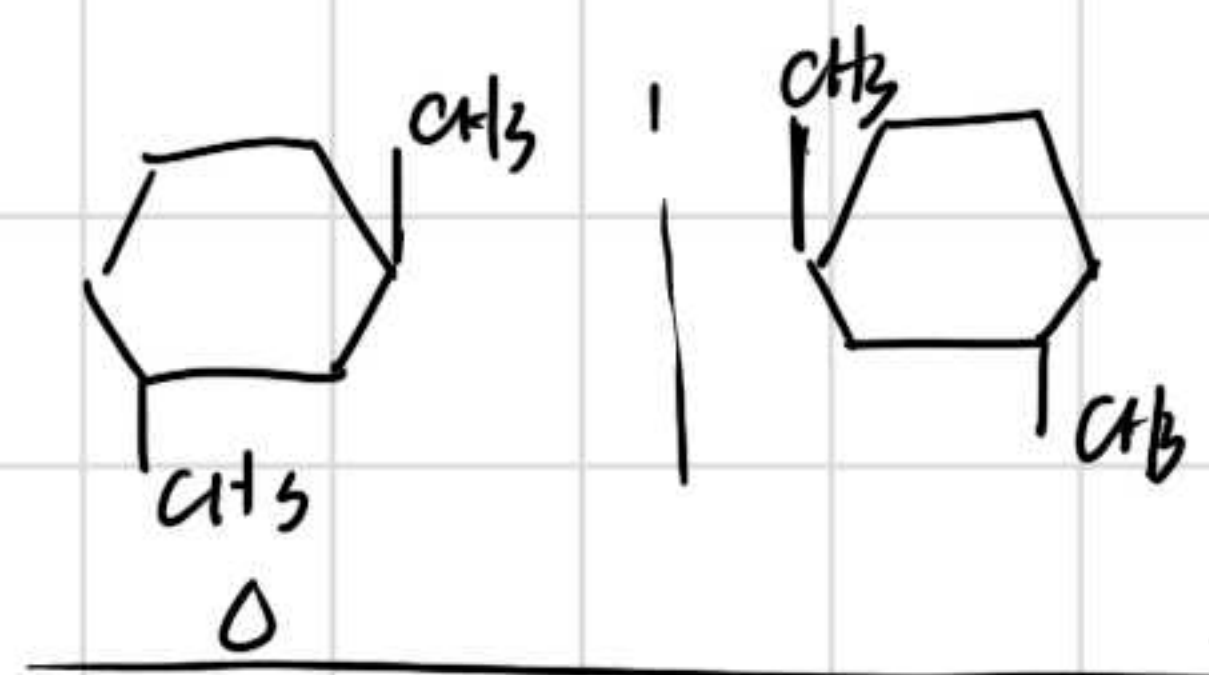
下列碳正离子中最不稳定的是( )



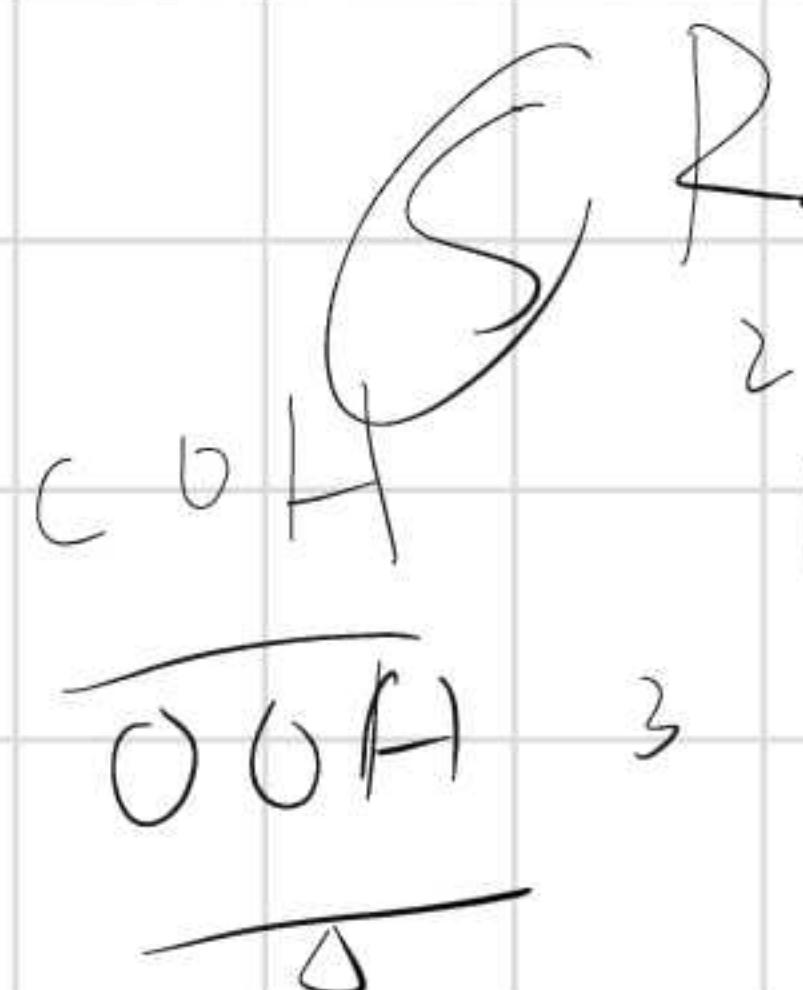
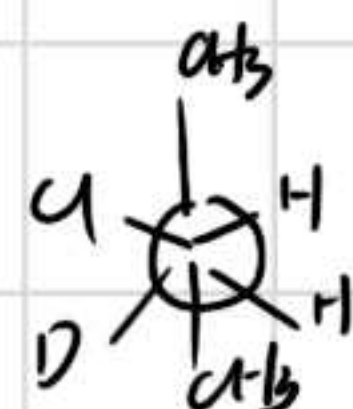
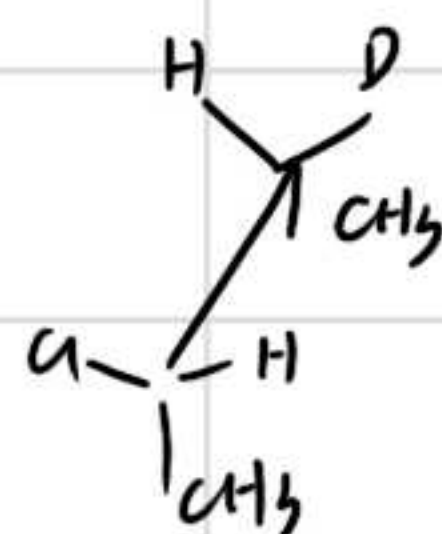
A | A

B | B

C | C



有利于!



16. 单选题(5.2 分) 中

甲苯比苯更容易发生芳环上的亲电取代反应, 主要原因是( )

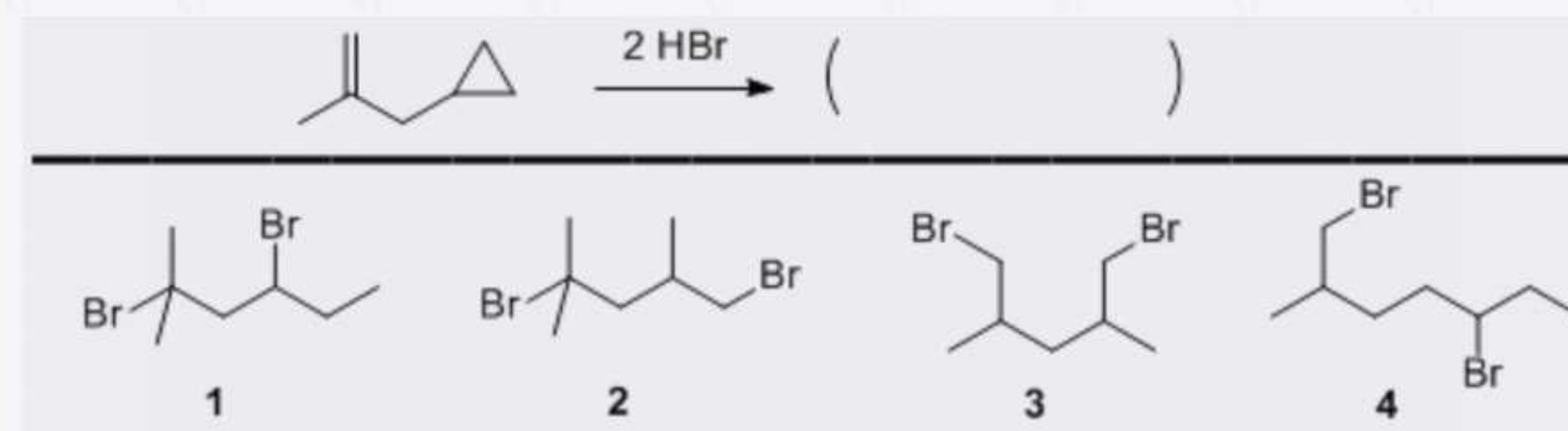
- A 甲基的诱导效应
- B 甲基的诱导效应以及甲基和苯之间的超共轭效应
- C 甲基和苯之间的超共轭效应

A | A

B | B

C | C

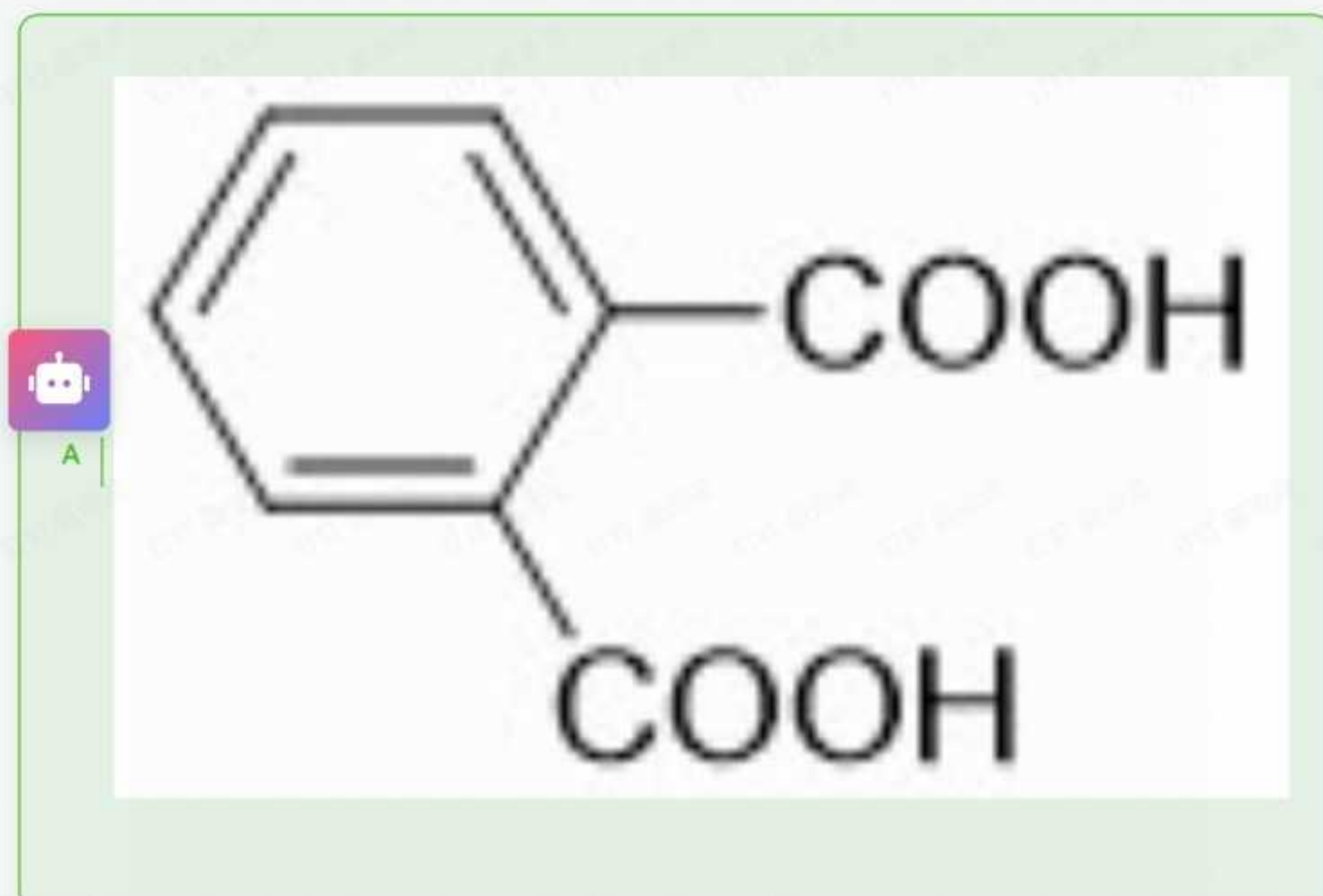
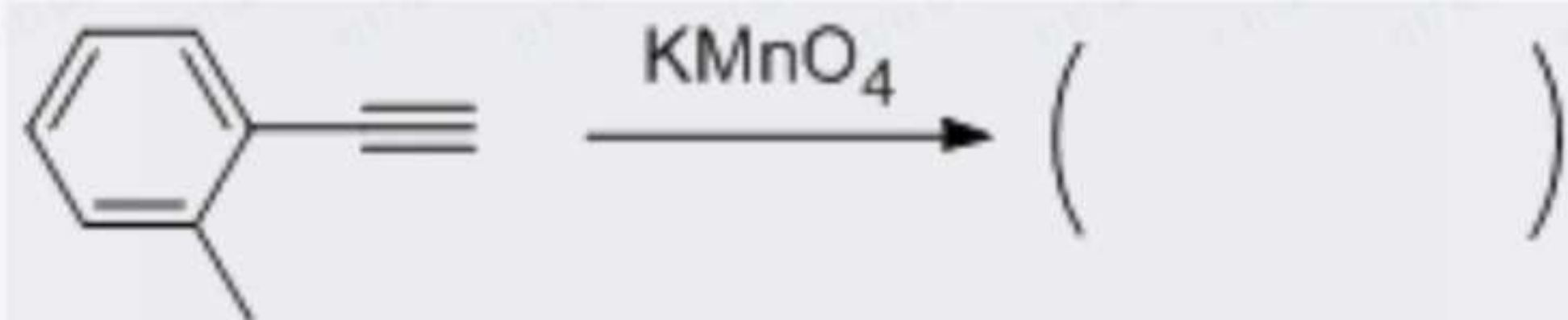
以下反应的主要产物是( )



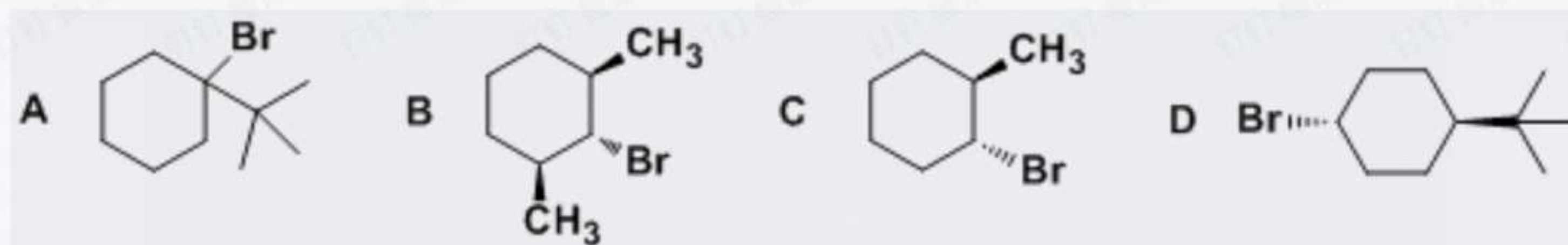
A | 1



选择以下反应的主要产物



下列化合物发生E2消去反应，速率最快的是( )



A | A

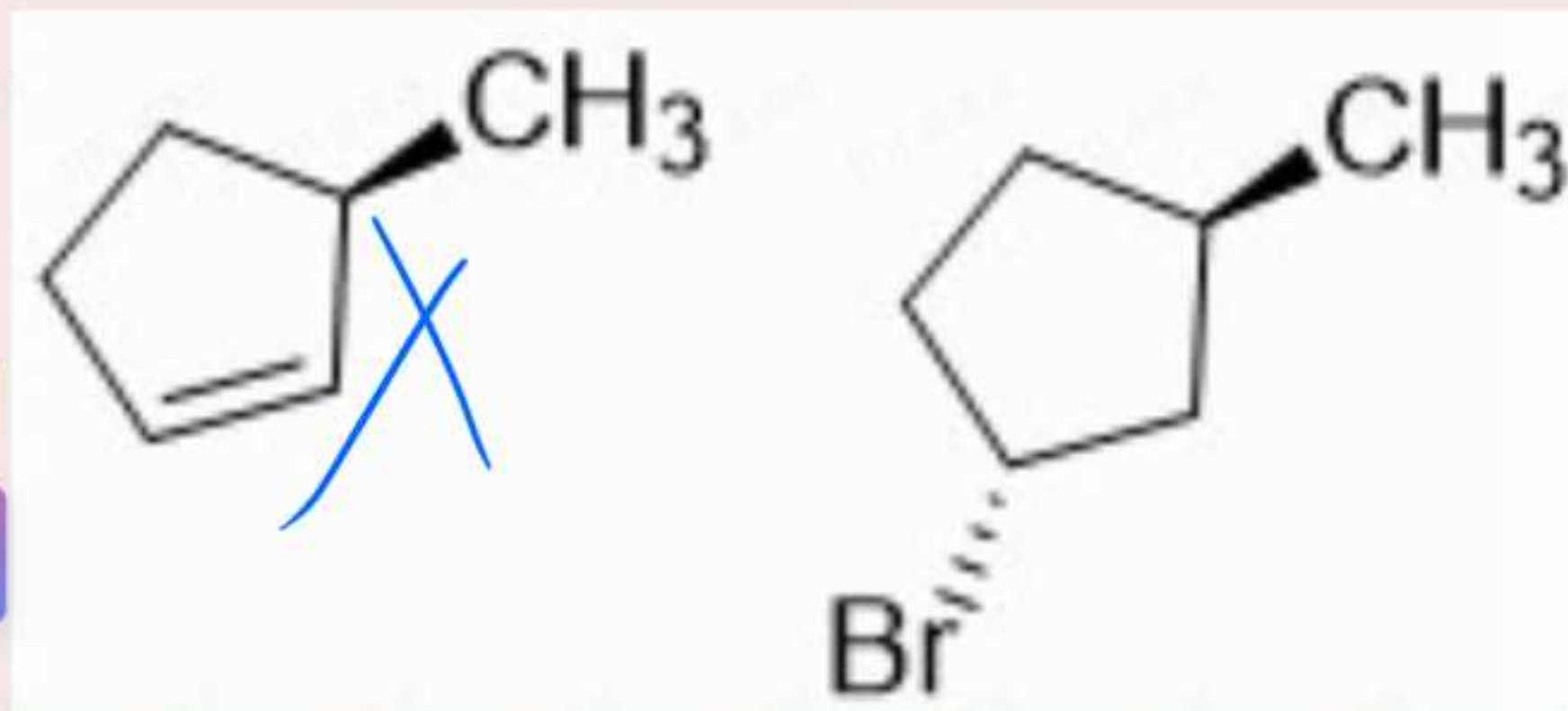
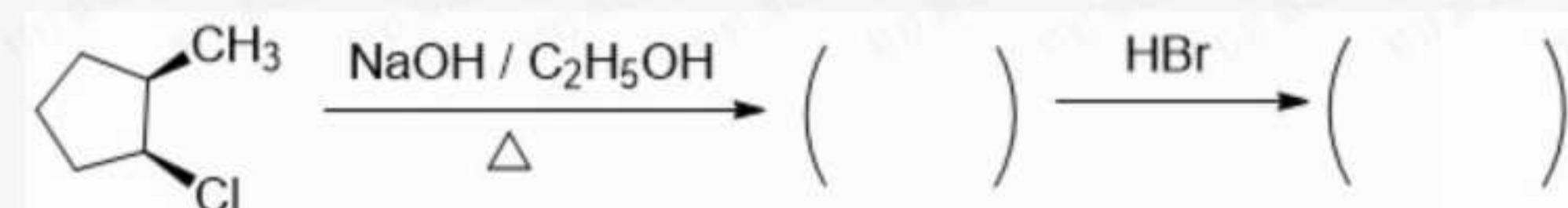
环丙烷( $C_3H_6$ ) 分子中C-C-C键角接近( )

A |  $109^\circ$

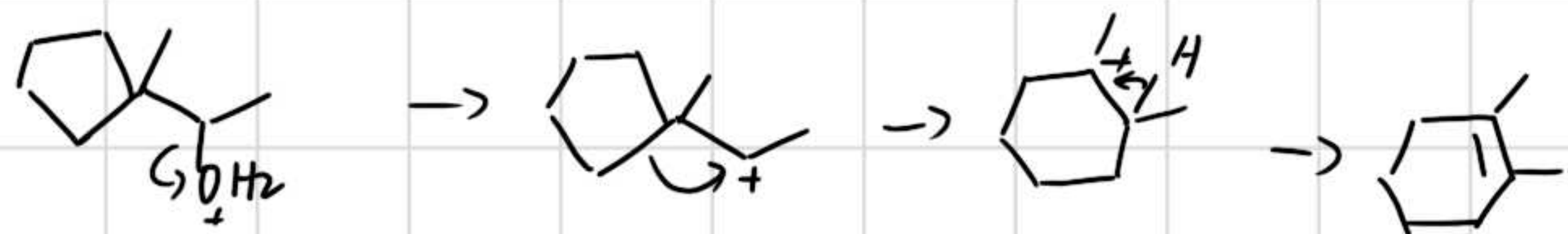
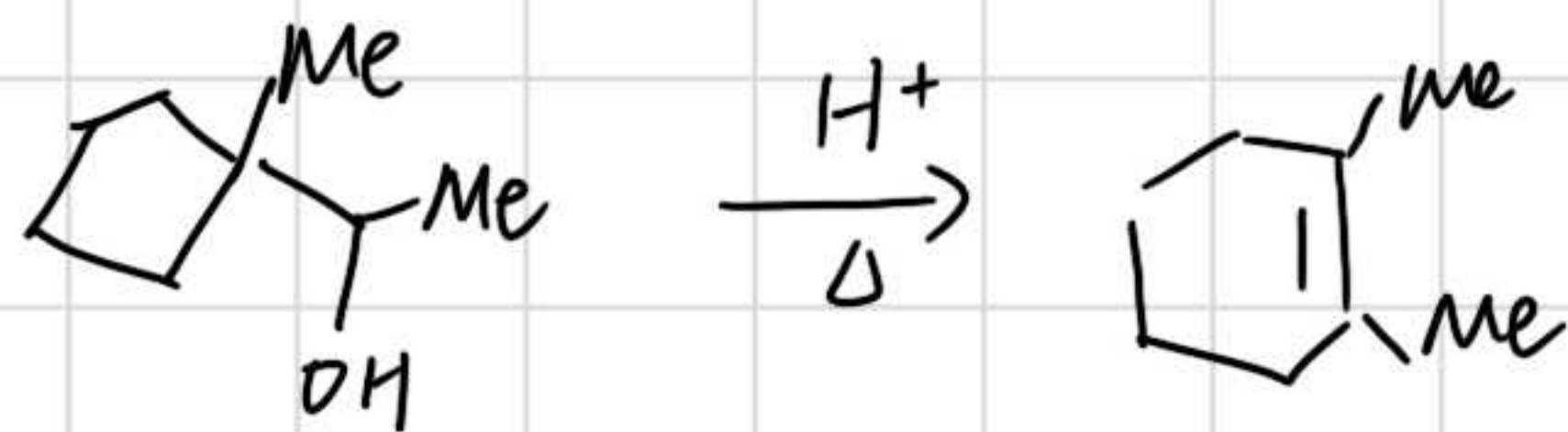
B |  $105^\circ$

C |  $120^\circ$

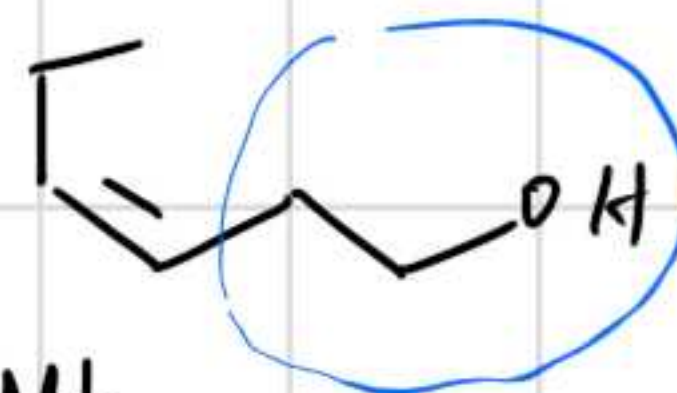
D |  $60^\circ$



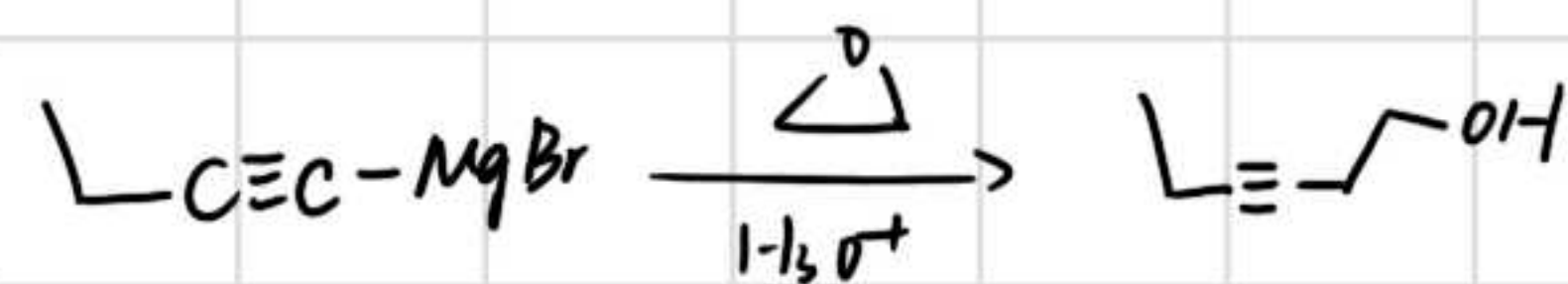
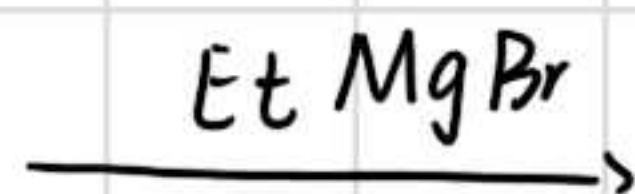
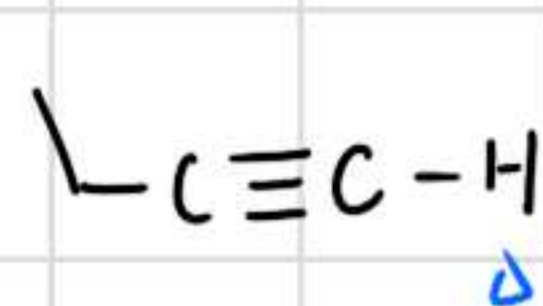
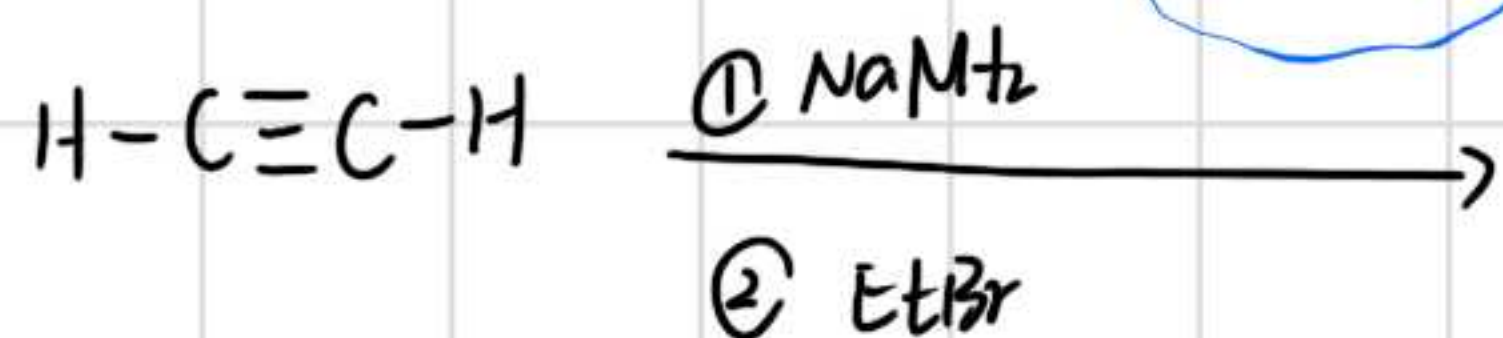




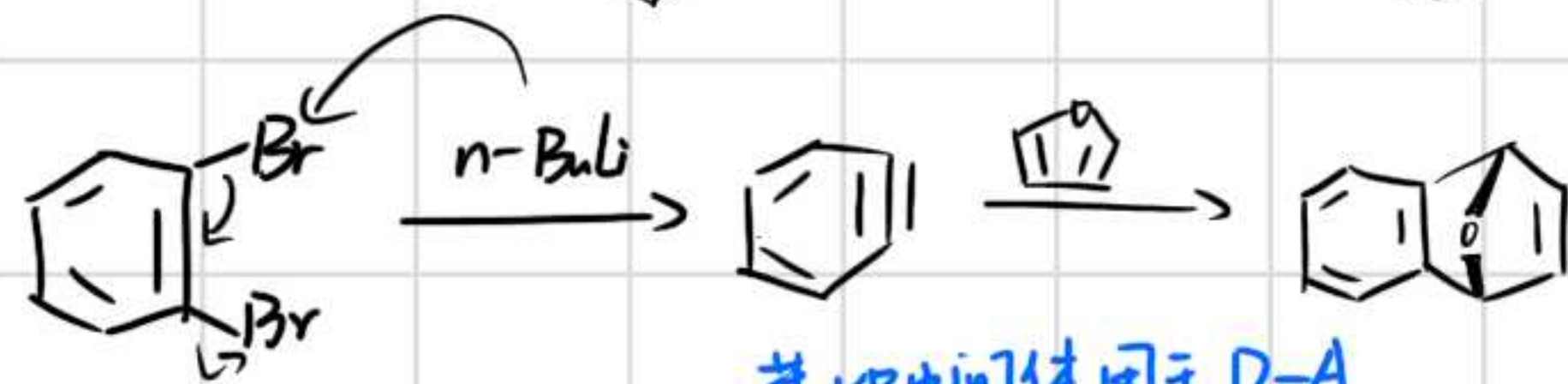
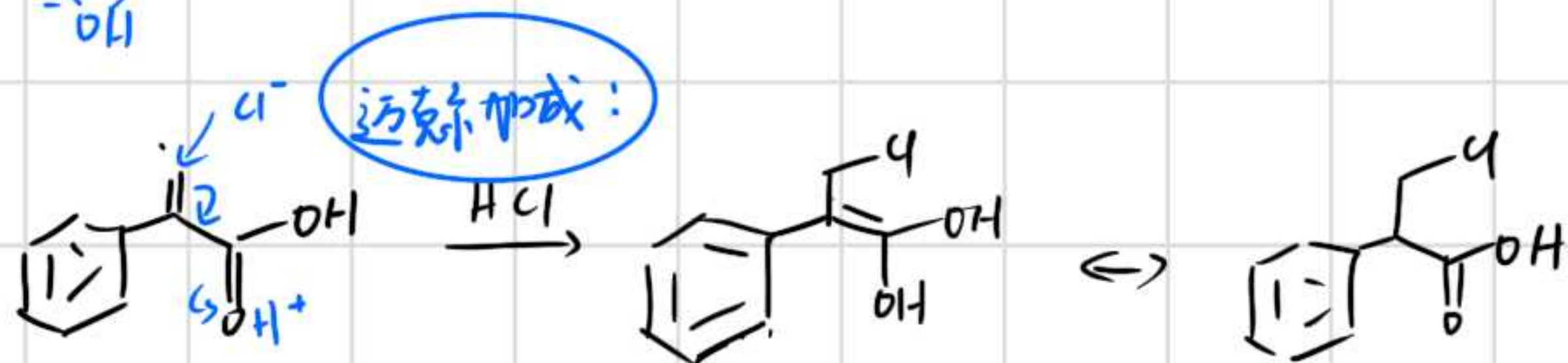
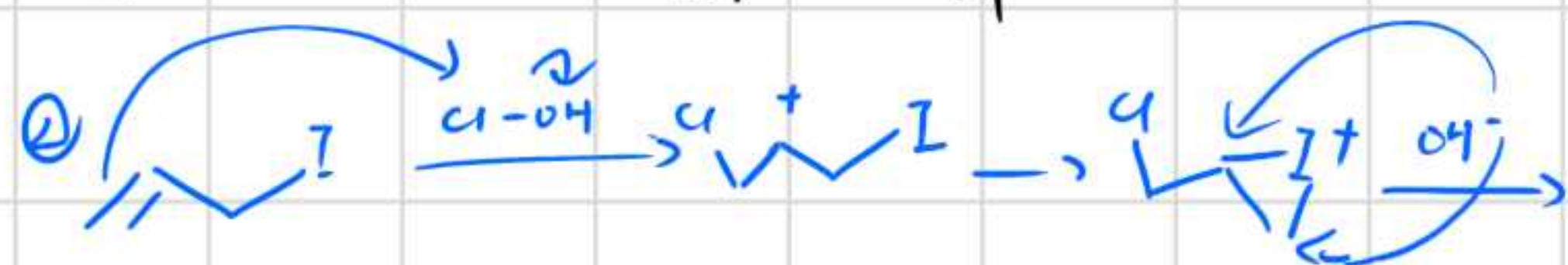
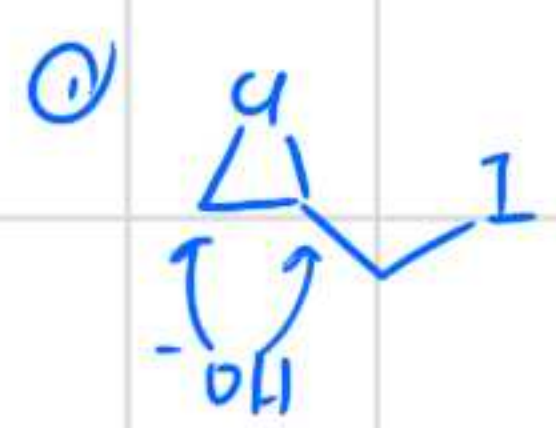
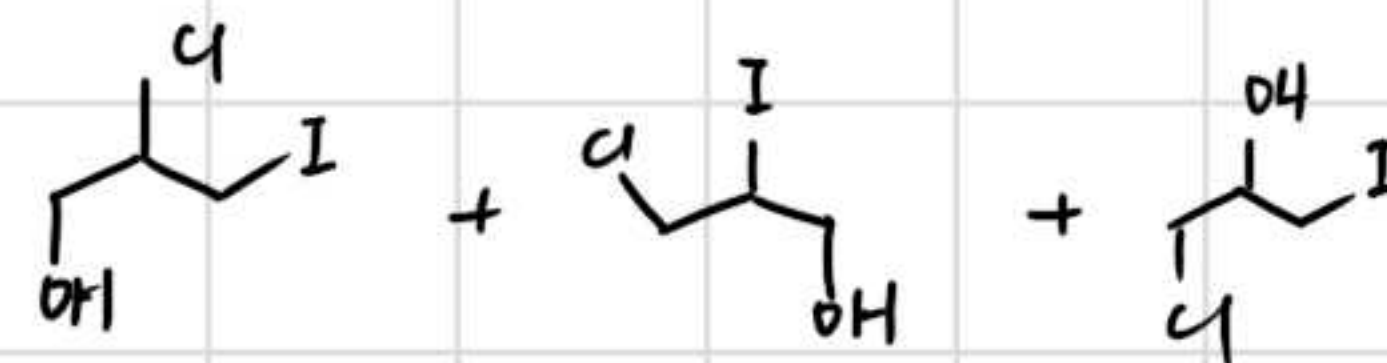
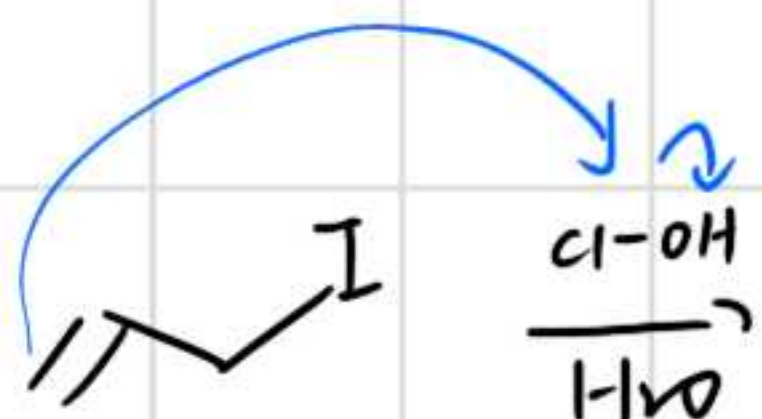
2. 炔起始合成



用环氧开环制备 进攻：用格氏试剂



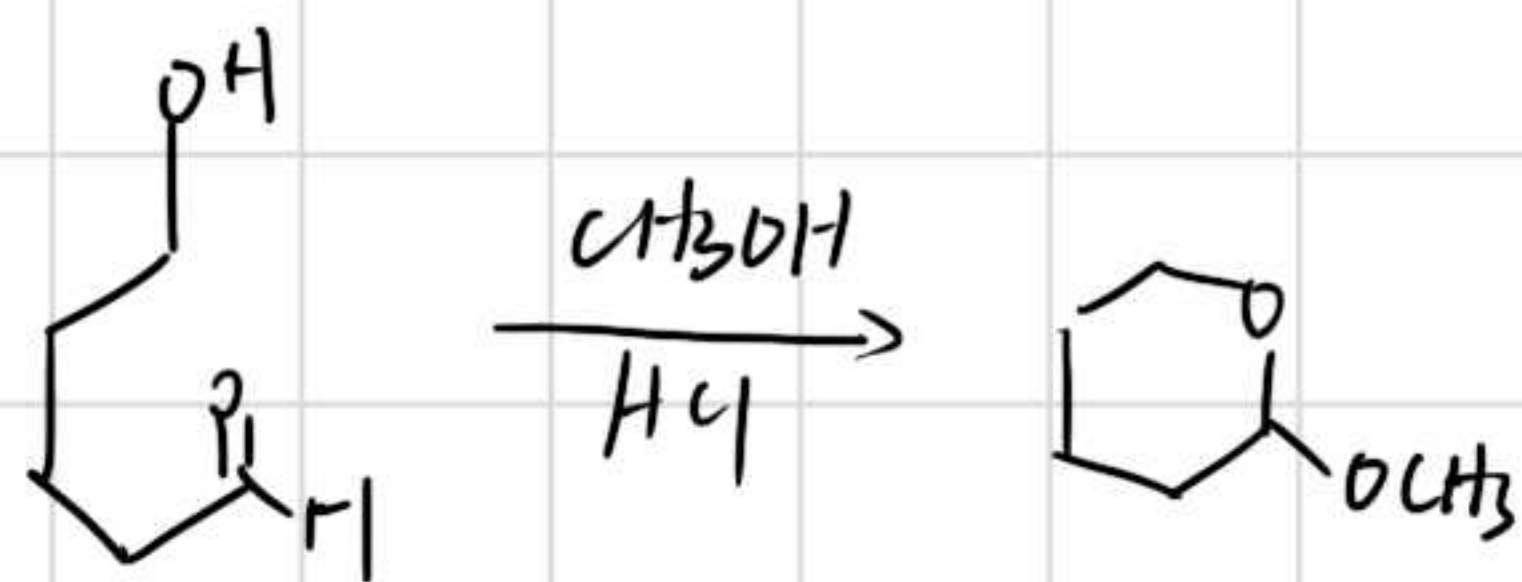
活泼H被拔：  
用于制备新格氏试剂



苯炔中间体用于 D-A

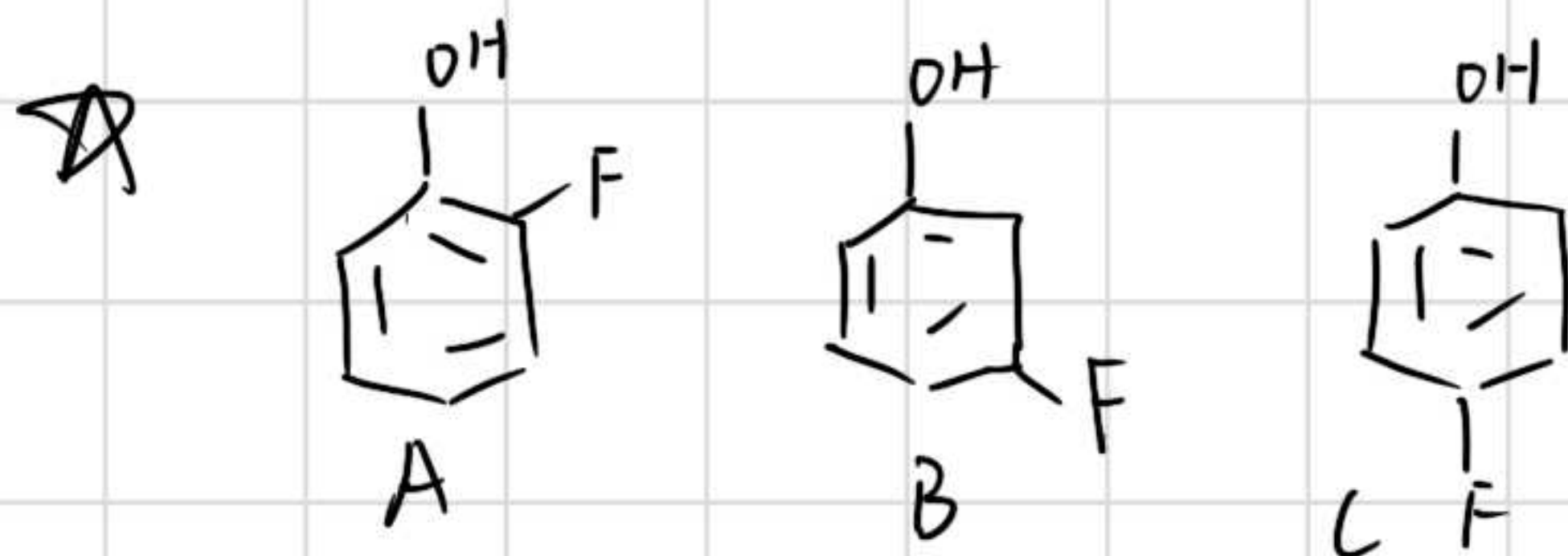
迈克尔加成：





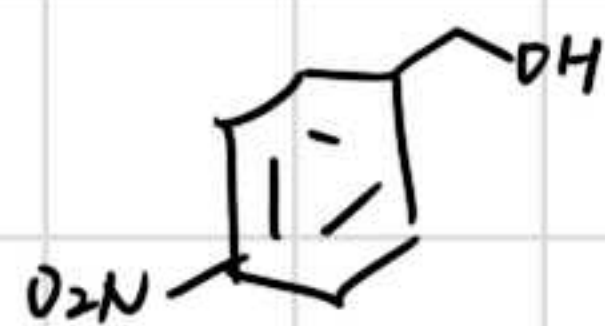
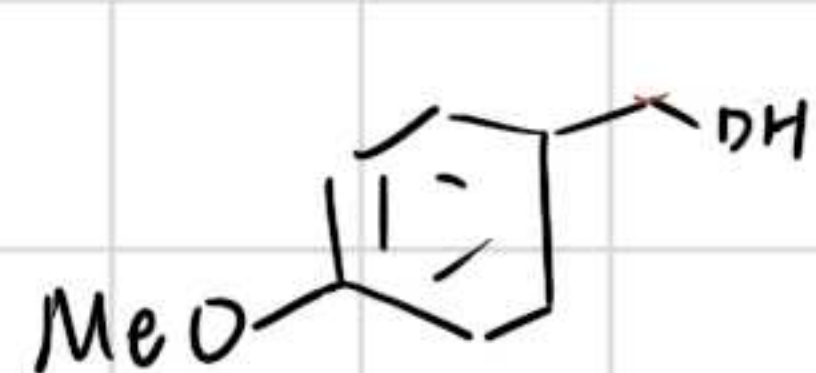
环内反应 变半缩醛

CH<sub>3</sub>OH 反应 变缩醛



酸性 A > B > C

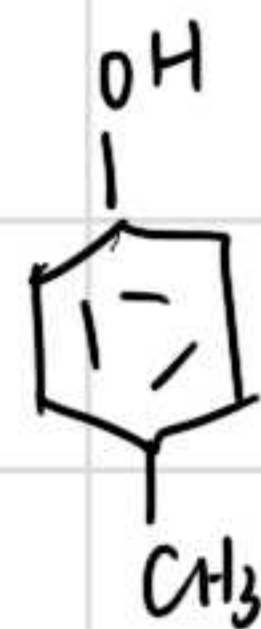
因素: 共轭给 + 诱导吸 考虑这个就行! 区别于 -NO<sub>2</sub>: 诱导吸 + 共轭吸



和 HBr 反应 V. ?

SN<sub>1</sub>

NaOH + 卤代烃的条件是酚氧负离子



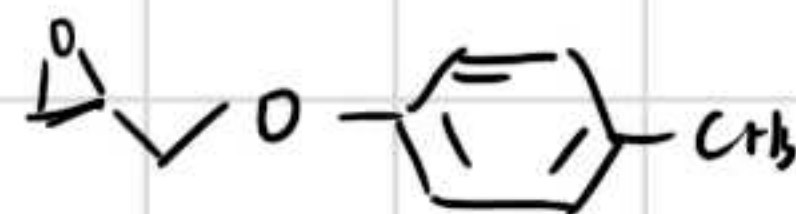
(1) NaOH

(2) CH<sub>3</sub>CH<sub>2</sub>Br

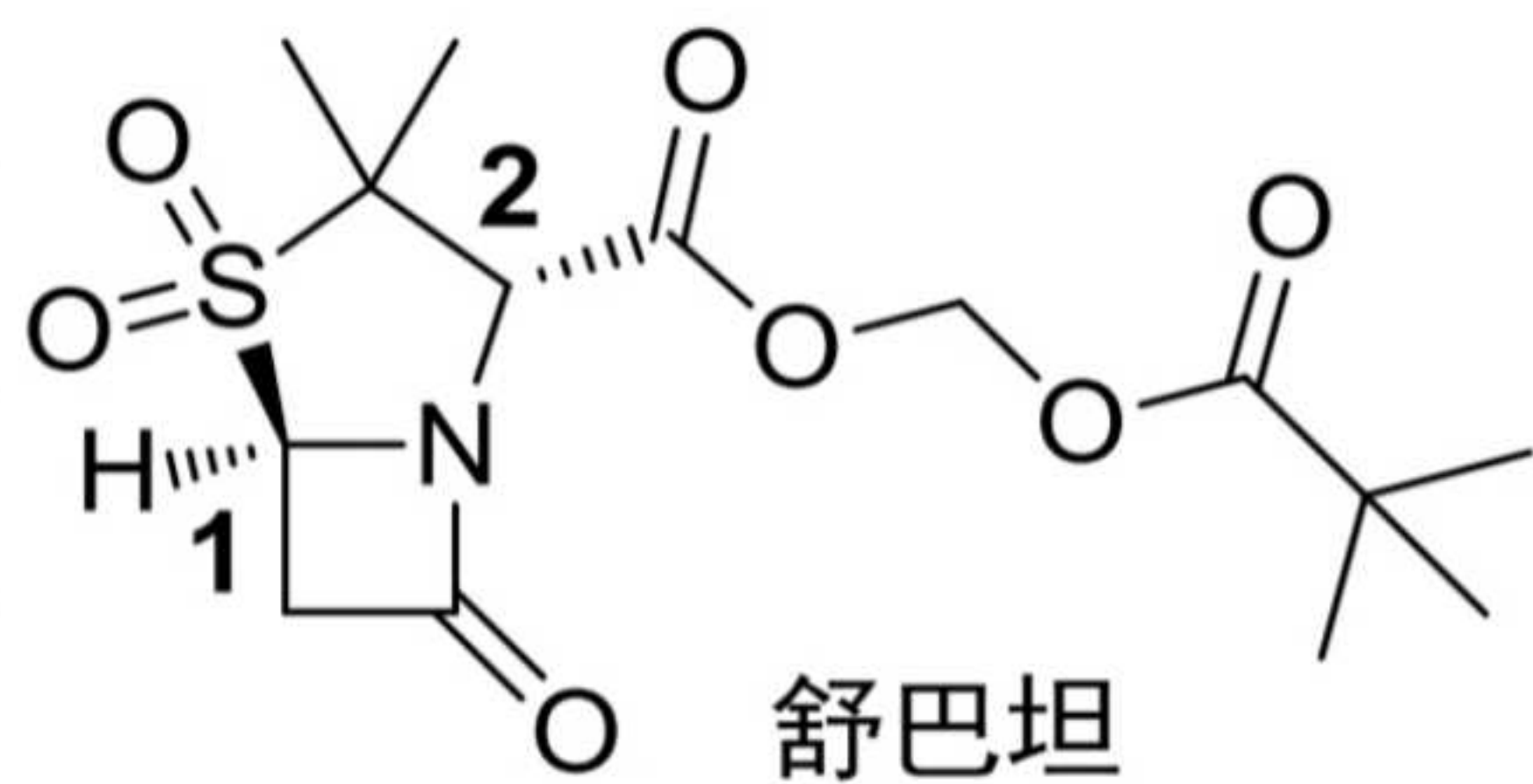
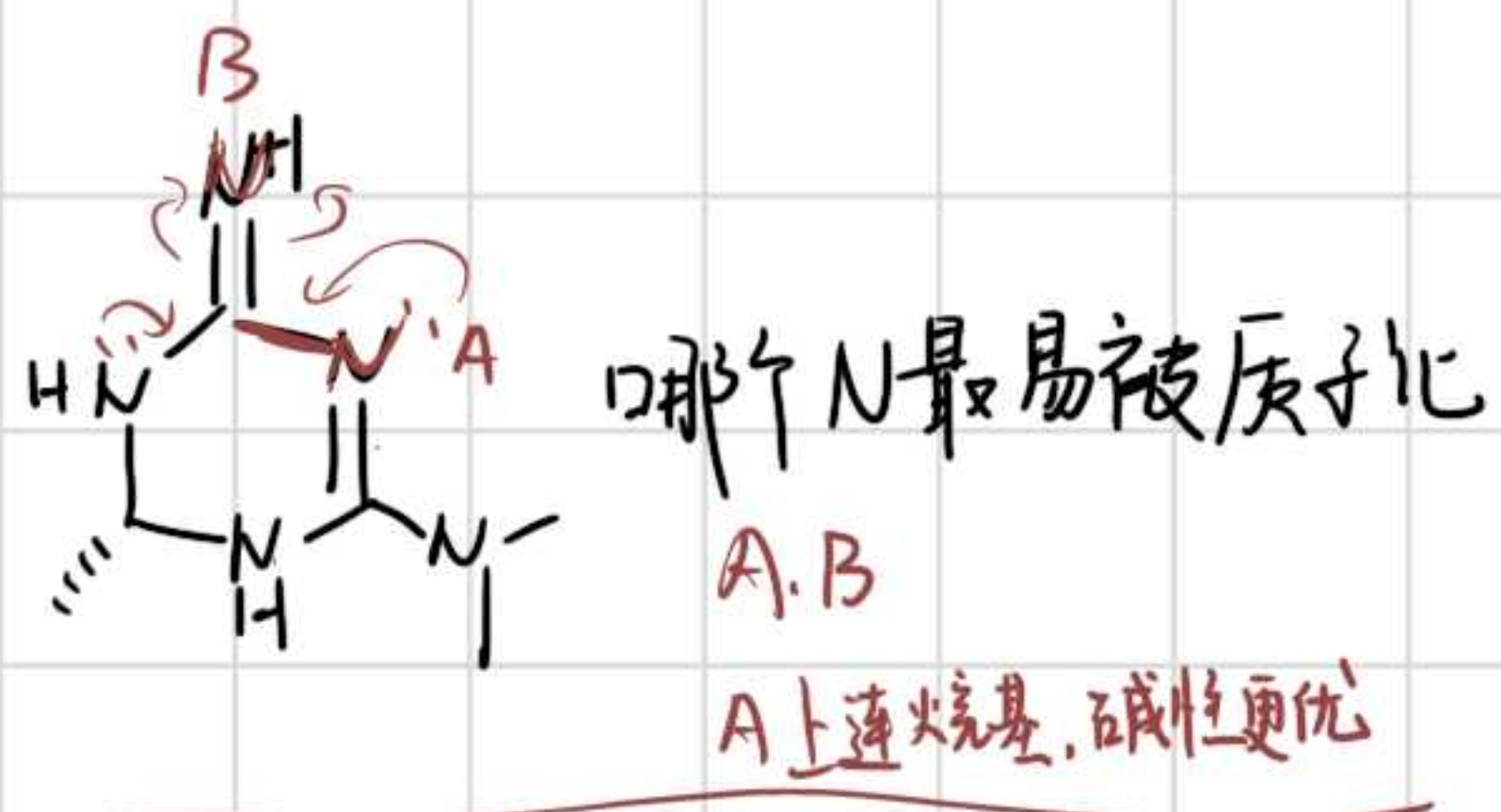


(1) NaOH

(2) ClCC1OC1







2 的构型:

易错!

R



